Prediction Based Dollar Cost Average Strategies: on Trump second Presidency

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Background and Motivation

Stock markets can be risky to invest in, President Trump’s tariffs make it even trickier. Economist Paul Krugman called it "unstable protectionism" (Morningstar, 2025).

Dollar-Cost Averaging (DCA) is able to manage risk, it allow investors to invest a fixed amount regularly, no matter what the market does.

What is Dollar Cost Average (DCA)?

Introduced by Benjamin Graham in The Intelligent Investor (1949).
It is a simple way to build wealth over time.
DCA is when you invest a fixed amount regularly, no matter what the market does.

DCA Example

Invest $100 every month

Month 1: Stock price $10 → Buy 10 shares

Month 2: Stock price $5 → Buy 20 shares

Month 3: Stock price $20 → Buy 5 shares

Total: $300 spent, 35 shares bought, average cost $8.57/share.

DCA +ve

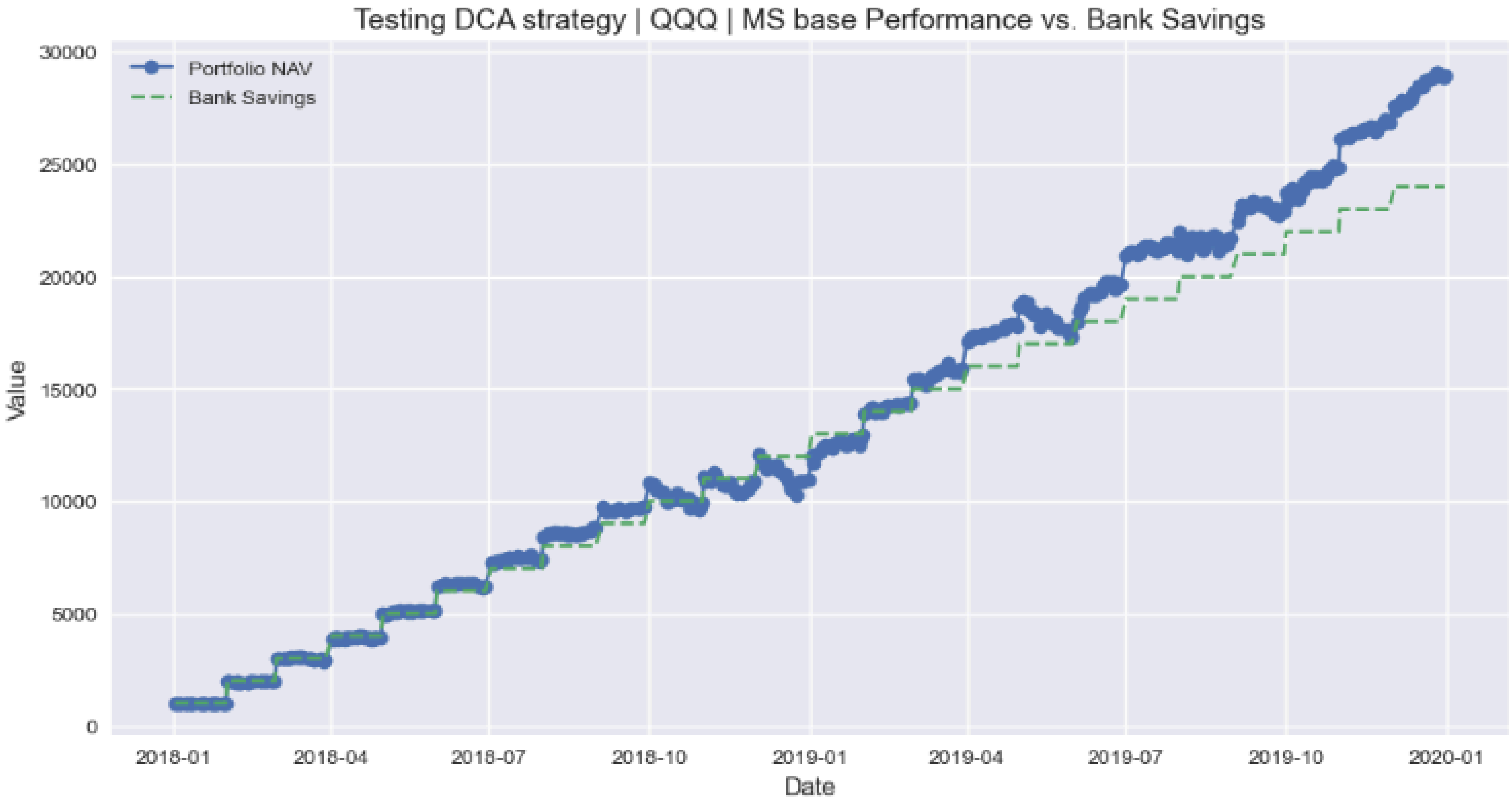
Straightforward, easy.

reduce stress of trying to predict market high and lows, avoid market emotional decision driven by fear or greed.

minimize the risk of poor timing decisions.

DCA result on Trump First Term

```
--------------------------------------------------------------------------
2019-12-31 | +++ CLOSING FINAL POSITION +++
2019-12-31 | Closing position of 140.0 share(s) at 205.88811
2019-12-31 | Current Balance: 28937.29
2019-12-31 | Net performance vs. bank savings (%) = 20.57
2019-12-31 | Total trades executed = 25
--------------------------------------------------------------------------
```



DCA -ve

Inflexible

- lack of adaptability, lead to underperforming in markets with strong trends on upward or downward, due to to its fixed investment schedule (Kirkby et al. 2020)
- underperform on lump-sum strategies by a large markets on backtesting during 2000 to 2010 (Kirkby et al. 2020)
- mainly due to the continued investments during prolonged declines, increasing losses. (Kirkby et al. 2020)

DCA make loss on continued investing during decline

```
----------------------------------------------------------------------------
2019-12-31 | +++ CLOSING FINAL POSITION +++
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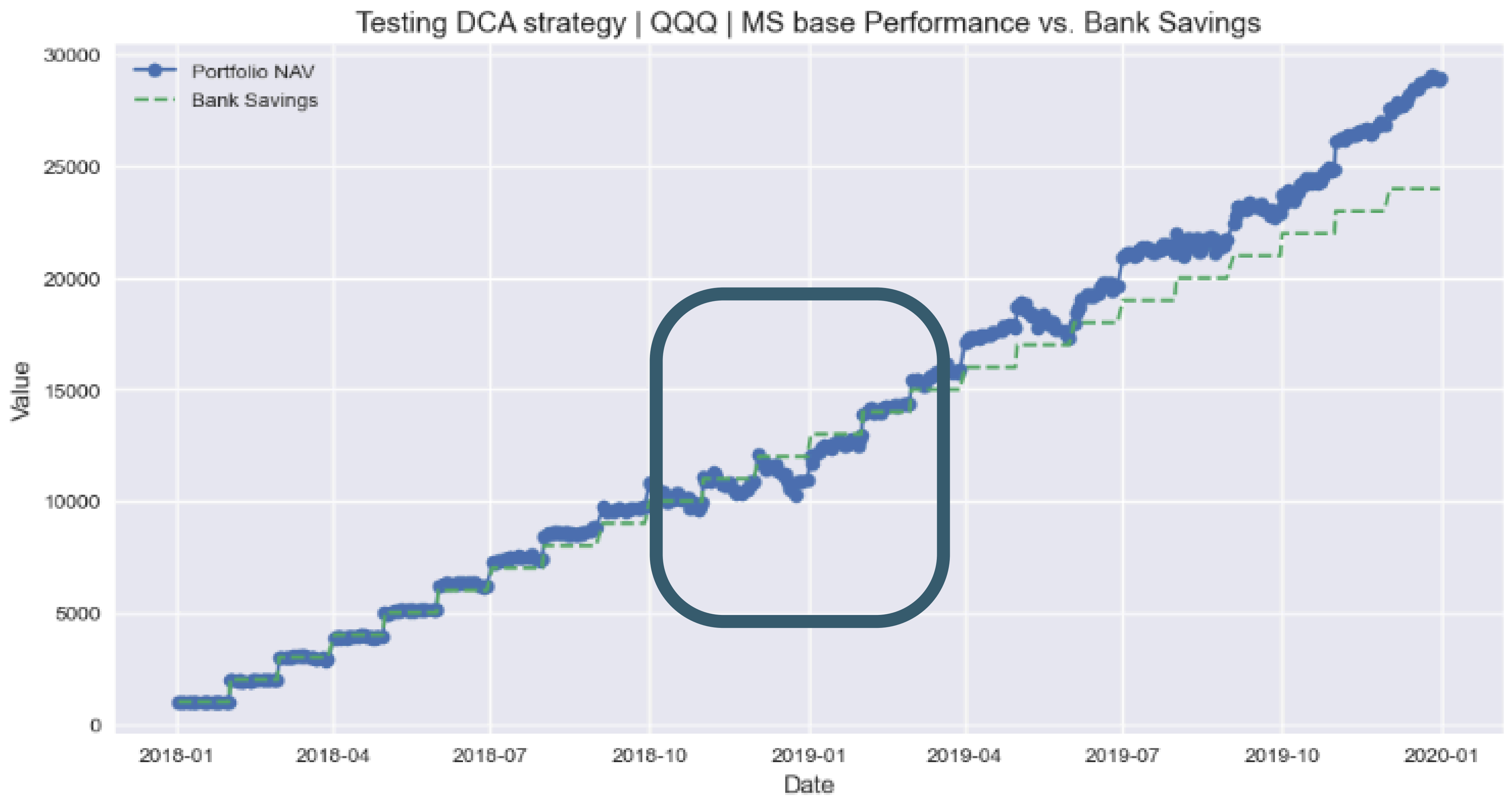


Motivation and Research Question

This research is aim to address the problem in continue invest a fixed amount of money to an asset.

Instead of a routine schedule, investment amounts are adjusted dynamically using regression model forecasts that draw on news sentiment, macroeconomic indicators, and technical signals.

DCA: Literature review

Tradtional DCA

Introduced by Benjamin Graham in The Intelligent Investor (1949). The great depression on 1929 led Graham to suggest fixed monthly investments to reduce losses (Graham, 1949).

In the 20th century, more people used in mutual funds to mitigate market risk. A notable example of disciplined, long-term investing is Warren Buffett, a student of Graham. Buffett's successfully outperforming active fund managers, highlights the power of patient investing (Bogle, 2007).

DCA:

Fixed **$100**

1st month

- Stock price $10 → Buy 10 shares

2nd month

- Stock price $5 → Buy **30** shares

3rd month

- Stock price $20 → Buy **5** shares

Enhanced DCA (EDCA)

Enhanced Dollar-Cost Averaging (EDCA), created by Dunham and Friesen (2012), adjusts investment amounts based on recent market performance, increasing allocations after market dips and reducing them after rallies.

Backtesting from 2000 to 2009 across stock indices and mutual funds showed that EDCA outperformed traditional DCA by 30 to 70 basis points annually by capitalizing on lower prices during downturns (Dunham & Friesen, 2012).

DCA:

Fixed **$100**

1st month
- Stock price $10 → Buy 10 shares

2nd month
- Stock price $5 → Buy **30** shares

3rd month
- Stock price $20 → Buy **5** shares

EDCA:

Not Fixed $100, return dec **$150** inc **$50**

1st month
- Stock price $10 → Buy 10 shares

2nd month
- Stock price **$5** → Buy **60** shares

3rd month
- Stock price **$20** → Buy **2.5** shares

Adaptive DCA (ADCA)

ADCA, introduced by Flanagan and Greenhut (2021), uses broader economic indicators—such as market volatility, unemployment rates, and capacity utilization—to dynamically adjust investment sizes.

Experiments showed that ADCA involved investing more during bear markets and less during bull markets, yielding superior risk-adjusted returns over traditional DCA in backtests from 1967 to 2018 (Flanagan & Greenhut, 2021).

ADCA:

Not Fixed, Ump rate dec **$150** inc **$50**

1st month
- Ump Rate 50 → Buy 10 shares

2nd month
- Ump Rate **80** → Buy **30** shares

3rd month
- Ump Rate **20** → Buy **5** shares

EDCA:

Not Fixed $100, return dec **$50** inc **$150**

1st month
- Stock price $10 → Buy 10 shares

2nd month
- Stock price **$5** → Buy **60** shares

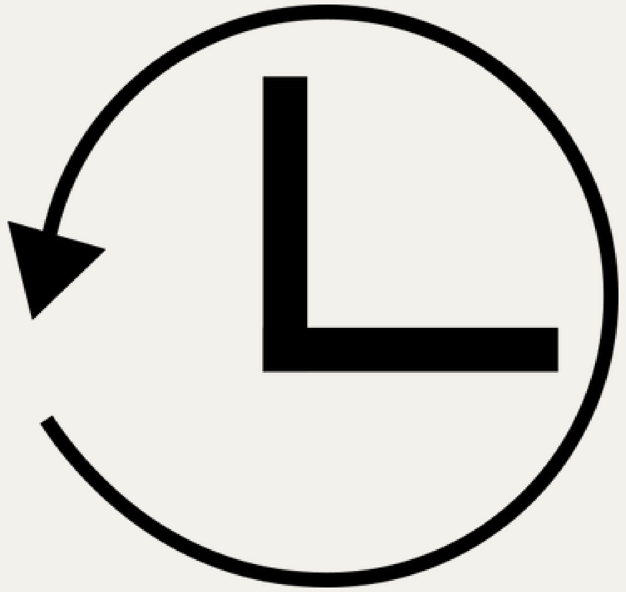
3rd month
- Stock price **$20** → Buy **2.5** shares

Limation on EDCA and ADCA

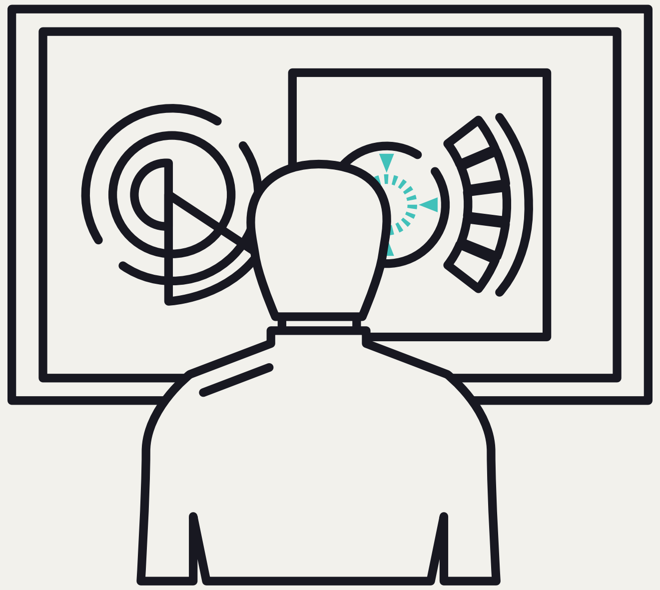
Rely on past data, lagging indicators, lead to delays in decisions (Investopedia, 2025)

- Economic or company shifts misjudged by lagging indicators
- EDCA sometimes invests more before drops cause losses (CXO Advisory, 2012)
- ADCA use slow indicators like unemployment may cause losses (AvaTrade, 2024)

lagging indicators v.s. Leading indicators



- lagging indicators give insight into past outcomes
- tell you about the health of an organization.



- Leading indicators help predict future performance
- tell you about the future movement

Machine Learning can be the solutions

Treat as leading indicators by forecast trends

- Identifies patterns in market behavior (Smith et al., 2023)
- Neural networks predict stock returns (Johnson & Lee, 2022)
- LSTM models forecast next-day closing prices (Chen et al., 2021)

Methodology: Database

Database Design: Collect Financial Data, and Create a Self-updating Stock Database

Collects and stores data from Yahoo Finance (stock prices), Finviz (news headlines), and investing.com (macroeconomic indicators)

Store on SQLite database

Price Data (QQQ)

Price Data on python

```
fetch_daily_data('QQQ', start='2025-04-01', end='2025-04-10')
```

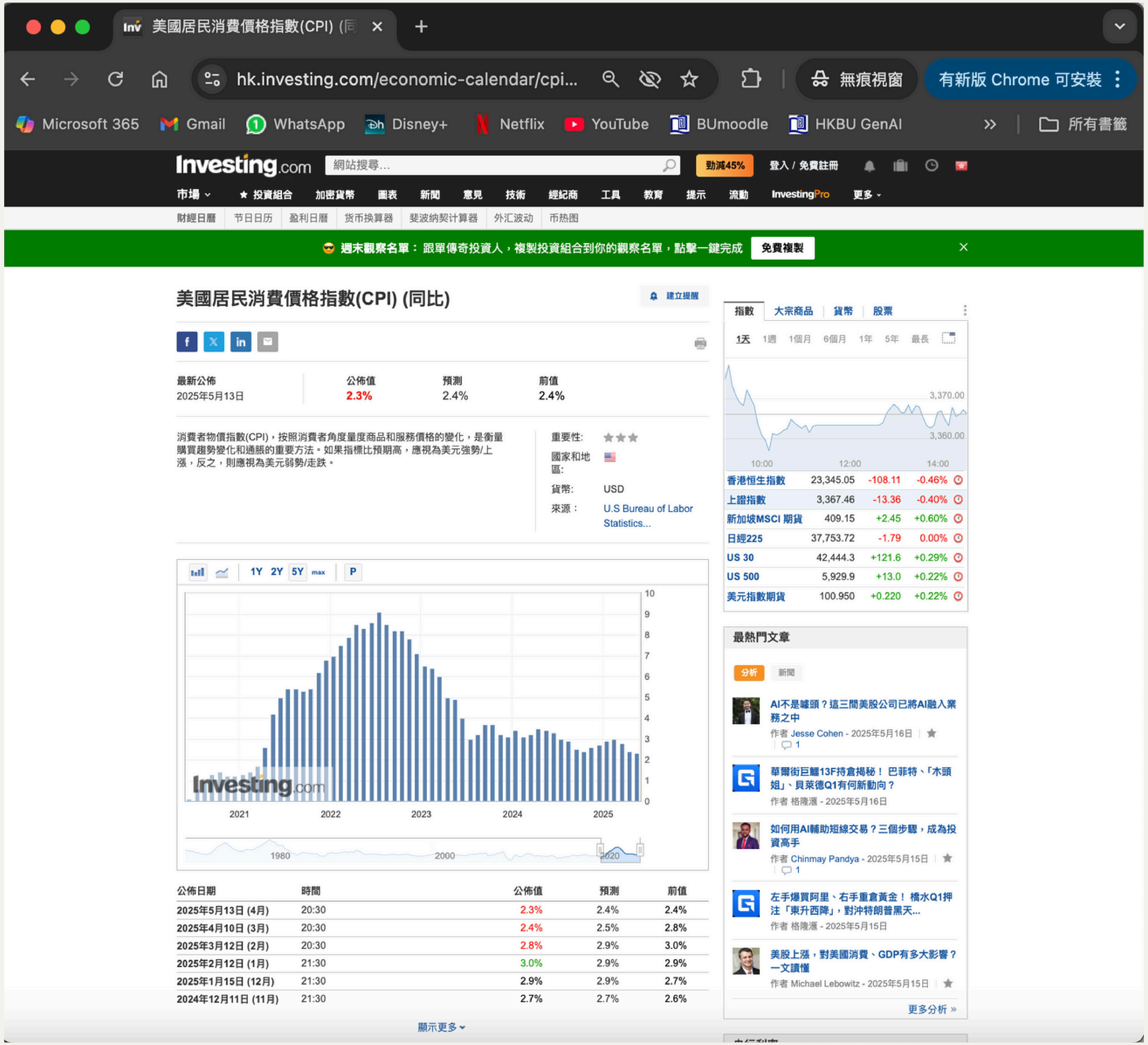
```
YF.download() has changed argument auto_adjust default to True
[*********************100%***********************]  1 of 1 completed
```

	date	open	high	low	close	volume	ticker	month_year
0	2025-04-01	467.299988	473.630005	464.420013	472.700012	41156200	QQQ	2025-04
1	2025-04-02	466.119995	479.559998	465.859985	476.149994	49894500	QQQ	2025-04
2	2025-04-03	456.440002	460.070007	450.140015	450.660004	70456300	QQQ	2025-04
3	2025-04-04	438.140015	440.369995	422.670013	422.670013	117088400	QQQ	2025-04
4	2025-04-07	408.660004	443.140015	402.390015	423.690002	161557000	QQQ	2025-04
5	2025-04-08	438.160004	443.140015	409.790009	416.059998	101248100	QQQ	2025-04
6	2025-04-09	415.570007	467.829987	415.429993	466.000000	142876900	QQQ	2025-04

Price Data Description

Date	This shows when the data was recorded, using the format year-month-day (YYYY-MM-DD).
Close	The price of the stock when trading ended on that day.
High	The highest price the stock reached on that day.
Low	The lowest price the stock reached on that day.
Open	The price at which the stock started trading on that day.
Volume	The total number of shares the stock were traded on that day.
month_year	Shows the month and year of the data record, formatted as YYYY-MM.

Economic Data (US)



Economic Data on python

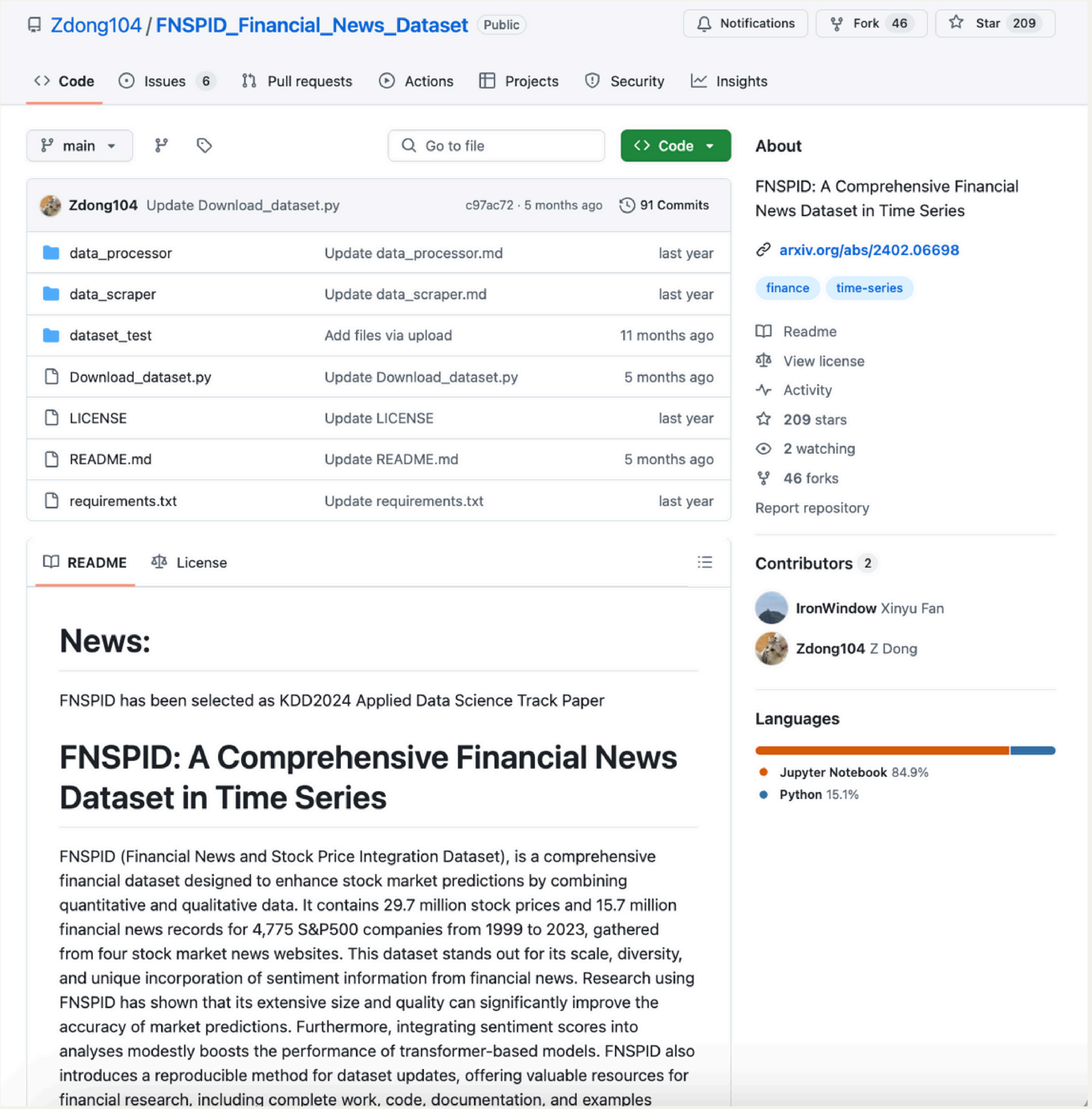
```
get_economic_data()
```

	month_year	Release_Date	Unemployment_Actual	Unemployment_Predicted	CPI_Actual	CPI_Predicted	Nonfarm_Pa
0	2025-05	2025-05-02	NaN	NaN	NaN	NaN	
1	2025-04	2025-04-04	0.042	0.041	0.024	0.025	
2	2025-03	2025-03-07	0.041	0.040	0.028	0.029	
3	2025-02	2025-02-07	0.040	0.041	0.030	0.029	
4	2025-01	2025-01-10	0.041	0.042	0.029	0.029	
5	2024-12	2024-12-06	0.042	0.042	0.027	0.027	

Economic Data Description

Unemployment_Actual	The real unemployment rate in the U.S., shown as a percentage.
Unemployment_Predicted	The forecasted unemployment rate in the U.S., shown as a percentage.
CPI_Actual	The real Consumer Price Index (CPI) value in the U.S., shown as a percentage.
CPI_Predicted	The forecasted CPI value in the U.S., shown as a percentage.
Nonfarm_Payrolls_Actual	The actual number of jobs added or lost in the U.S., excluding farm jobs.
Nonfarm_Payrolls_Predicted	The forecasted number of jobs added or lost in the U.S., excluding farm jobs.
Retail_Sales_Actual	The real total value of retail sales in the U.S.
Retail_Sales_Predicted	The forecasted total value of retail sales in the U.S.
Industrial_Production_Actual	The real index value measuring manufacturing and mining output in the U.S., shown as a percentage.
Industrial_Production_Predicted	The forecasted index value measuring manufacturing and mining output.
Consumer_Confidence_Index_Actual	The real index value showing how confident consumers are about the economy.
Consumer_Confidence_Index_Predicted	The predicted consumer confidence index value for the relevant market or region.
Personal_Income_Actual	The real total income received by individuals in the U.S.
Personal_Income_Predicted	The forecasted total income received by individuals in the U.S.

Historical News (FNSPID)

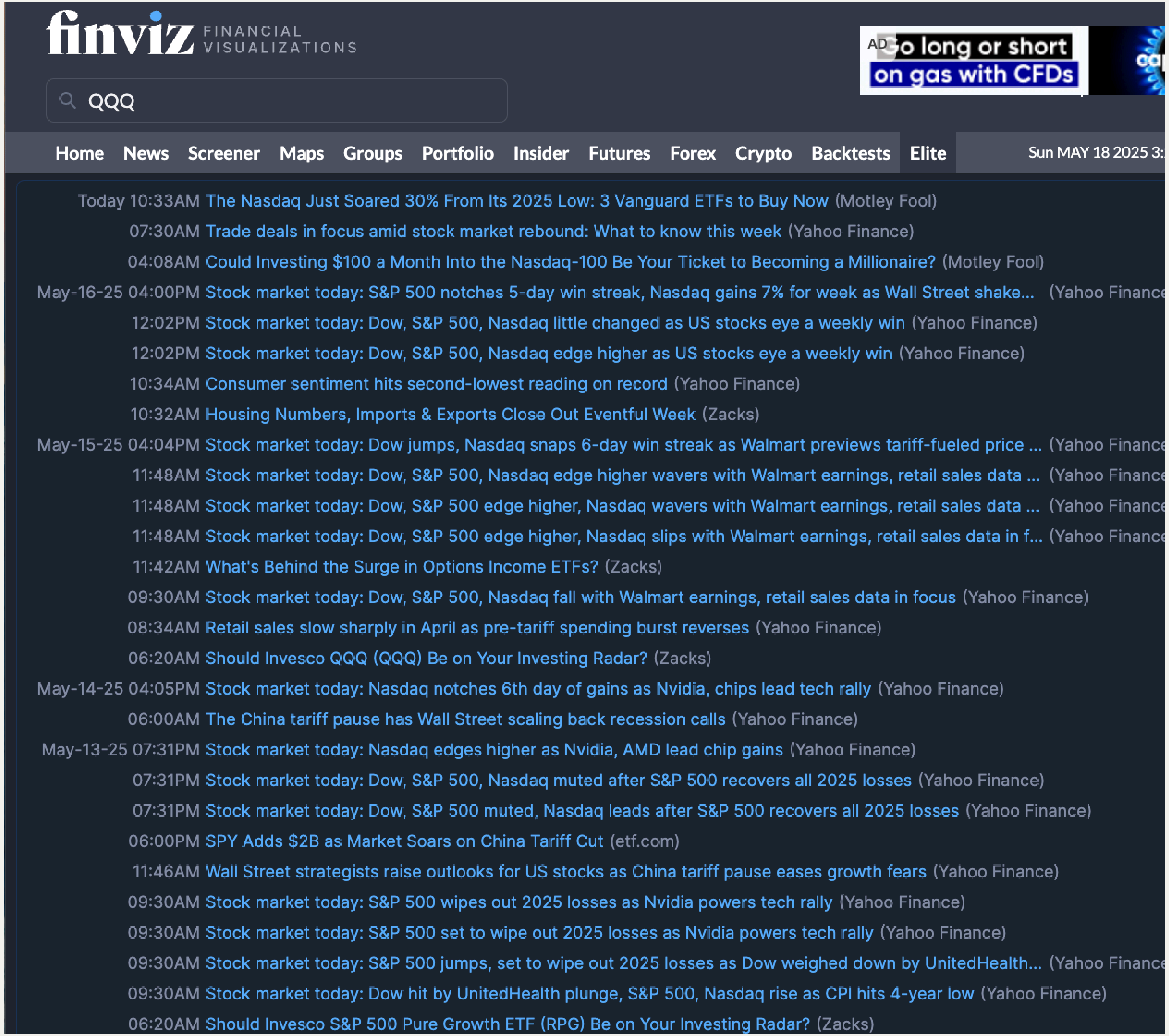


Historical News on python

```
historical_news_df = pd.read_csv('/Users/kevinleungch421/Documents/GitHub/Database/nasdaq_exteral_data')
historical_news_df.head()
```

	Date	Article_title	Stock_symbol	month_year
0	2023-12-16	Building Permits, Unemployment Claims and Othe...	QQQ	2023-12
1	2023-12-16	2 ETFs I Like More Than Invesco QQQ Trust for ...	QQQ	2023-12
2	2023-12-15	A Guide to Nasdaq ETF Investing	QQQ	2023-12
3	2023-12-15	Pre-Markets Take Profits to Conclude Momentous...	QQQ	2023-12
4	2023-12-15	How the Invesco QQQ ETF Could Turn You Into a ...	QQQ	2023-12

Recent News (Finzi)

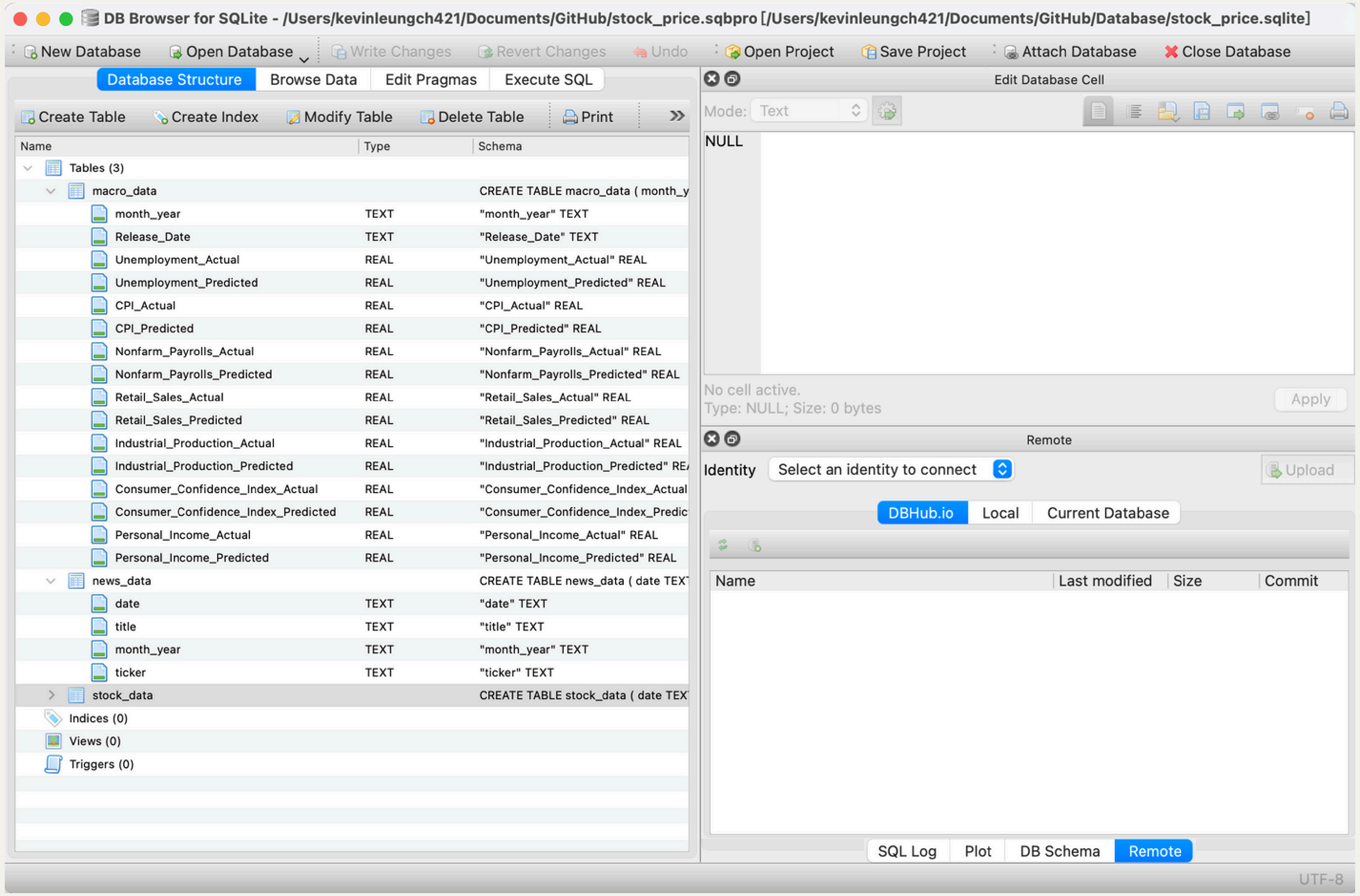


Recent News on python

```
news = get_news_data('QQQ', start='2025-04-01', end='2025-04-10')
news.head()
```

	date	title	month_year
45	2025-04-10	Consumer Price Index: Inflation Cools To 2.4% ...	2025-04
46	2025-04-10	Optimism Burns Off On Reentry To Economic Doub...	2025-04
47	2025-04-10	QQQ Pulls In $1.9 Billion of New Money During ...	2025-04
48	2025-04-10	March Inflation Surprises, And More Surprises ...	2025-04
49	2025-04-10	Stocks pummeled as Nasdaq sinks 4%, Dow drops ...	2025-04

Store on the SQLite DB browser



Schedule on update everyday

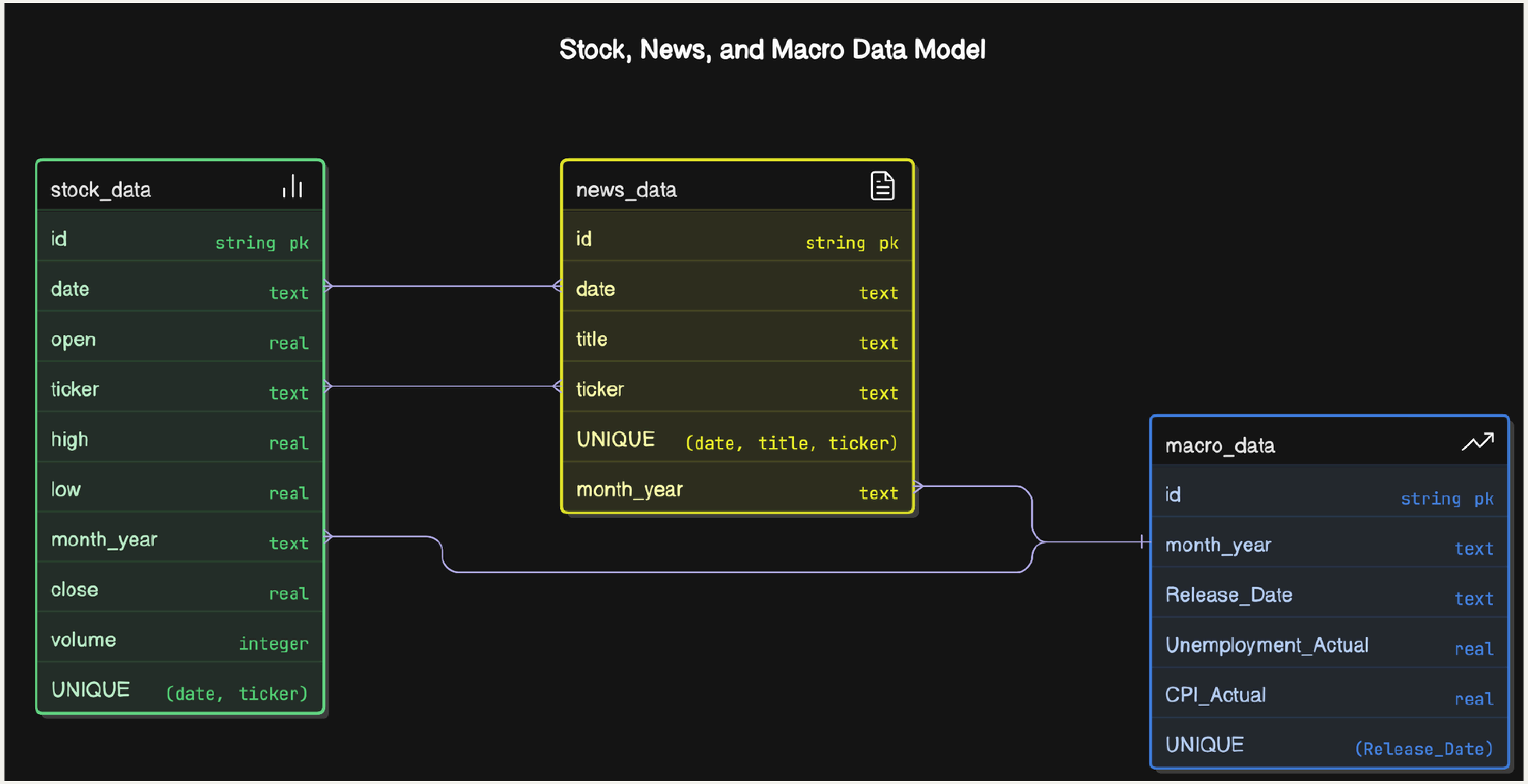
```
# # Import required libraries
import schedule
import time

# Assuming conn is already defined earlier in your script
conn = sqlite3.connect('stock_price.sqlite')

# Schedule daily updates at 9:00 AM
schedule.every().day.at("09:00").do(lambda: last_trade_date('QQQ', conn))       # Updates stock data
schedule.every().day.at("09:00").do(lambda: last_trade_date_news('QQQ', conn))  # Updates news data
schedule.every().day.at("09:00").do(lambda: last_trade_date_macro(conn))        # Updates macroeconomic d

# Run the scheduler continuously
while True:
    schedule.run_pending()
    time.sleep(1)
```

ER diagram


Stock, News, and Macro Data Model
stock_data
id string pk
date text
open real
ticker text
high real
low real
month_year text
close real
volume integer
UNIQUE (date, ticker)
news_data
id string pk
date text
title text
ticker text
UNIQUE (date, title, ticker)
month_year text
macro_data
id string pk
month_year text
Release_Date text
Unemployment_Actual real
CPI_Actual real
UNIQUE (Release_Date)

Methodology: Preprocess

Situation on Dataset

Dataset Rows & Columns count

```
print(f"Number of Rows: {trump_term_data.shape[0]}")
print(f"Number of Columns: {trump_term_data.shape[1]}")
```

```
Number of Rows: 2333
Number of Columns: 24
```

Dataset Information

```
trump_term_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2333 entries, 0 to 2332
Data columns (total 24 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   date                               2333 non-null   object
 1   open                               2333 non-null   float64
 2   high                               2333 non-null   float64
 3   low                                2333 non-null   float64
 4   close                              2333 non-null   float64
 5   volume                             2333 non-null   int64
 6   ticker                             2333 non-null   object
 7   month_year                         2333 non-null   object
 8   Release_Date                       2038 non-null   object
 9   Unemployment_Actual                2038 non-null   float64
 10  Unemployment_Predicted             2038 non-null   float64
 11  CPI_Actual                         2038 non-null   float64
 12  CPI_Predicted                      2038 non-null   float64
 13  Nonfarm_Payrolls_Actual            2038 non-null   float64
 14  Nonfarm_Payrolls_Predicted         2038 non-null   float64
 15  Retail_Sales_Actual                2029 non-null   float64
 16  Retail_Sales_Predicted             2038 non-null   float64
 17  Industrial_Production_Actual       2029 non-null   float64
 18  Industrial_Production_Predicted    2038 non-null   float64
 19  Consumer_Confidence_Index_Actual   2038 non-null   float64
 20  Consumer_Confidence_Index_Predicted 2038 non-null  float64
 21  Personal_Income_Actual             2029 non-null   float64
 22  Personal_Income_Predicted          2029 non-null   float64
 23  titles                             1866 non-null   object
dtypes: float64(18), int64(1), object(5)
memory usage: 437.6+ KB
```

Duplicate Values

```
trump_term_data.duplicated().sum()
```

```
0
```

Missing Values/Null Values

```
trump_term_data.isnull().sum()
```

```
date                                    0
open                                    0
high                                    0
low                                     0
close                                   0
volume                                  0
ticker                                  0
month_year                              0
Release_Date                          295
Unemployment_Actual                   295
Unemployment_Predicted                295
CPI_Actual                            295
CPI_Predicted                         295
Nonfarm_Payrolls_Actual               295
Nonfarm_Payrolls_Predicted            295
Retail_Sales_Actual                   304
Retail_Sales_Predicted                295
Industrial_Production_Actual          304
Industrial_Production_Predicted       295
Consumer_Confidence_Index_Actual      295
Consumer_Confidence_Index_Predicted   295
Personal_Income_Actual                304
Personal_Income_Predicted             304
titles                                467
dtype: int64
```

Check Unique Values for ea

```
for column in trump_term_data.columns:
    print(f"{column} : {len(trump_term_data[
```

```
date : 2333
open : 2332
high : 2332
low : 2333
close : 2299
volume : 2332
ticker : 1
month_year : 112
Release_Date : 99
Unemployment_Actual : 34
Unemployment_Predicted : 33
CPI_Actual : 55
CPI_Predicted : 53
Nonfarm_Payrolls_Actual : 86
Nonfarm_Payrolls_Predicted : 64
Retail_Sales_Actual : 41
Retail_Sales_Predicted : 30
Industrial_Production_Actual : 32
Industrial_Production_Predicted : 21
Consumer_Confidence_Index_Actual : 90
Consumer_Confidence_Index_Predicted : 79
Personal_Income_Actual : 26
Personal_Income_Predicted : 20
titles : 1860
```

Issue

- No Target variable
- News with noise, also need to transform into numeric data
- No prove that Trump 1st term can predict 2nd term
- No prove that DCA work
- lack of indicators
- Null value

1. Add target: Future Log Return

- This variable is important because it helps us understand how the price of QQQ is expected to change over the next month.

Formula:

$$\text{Future Log Return} = \ln\left(\frac{P_{\text{future}}}{P_{\text{current}}}\right)$$

Distribution on target variable

Issue

- ~~No Target variable~~
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2. General Cleaning on News

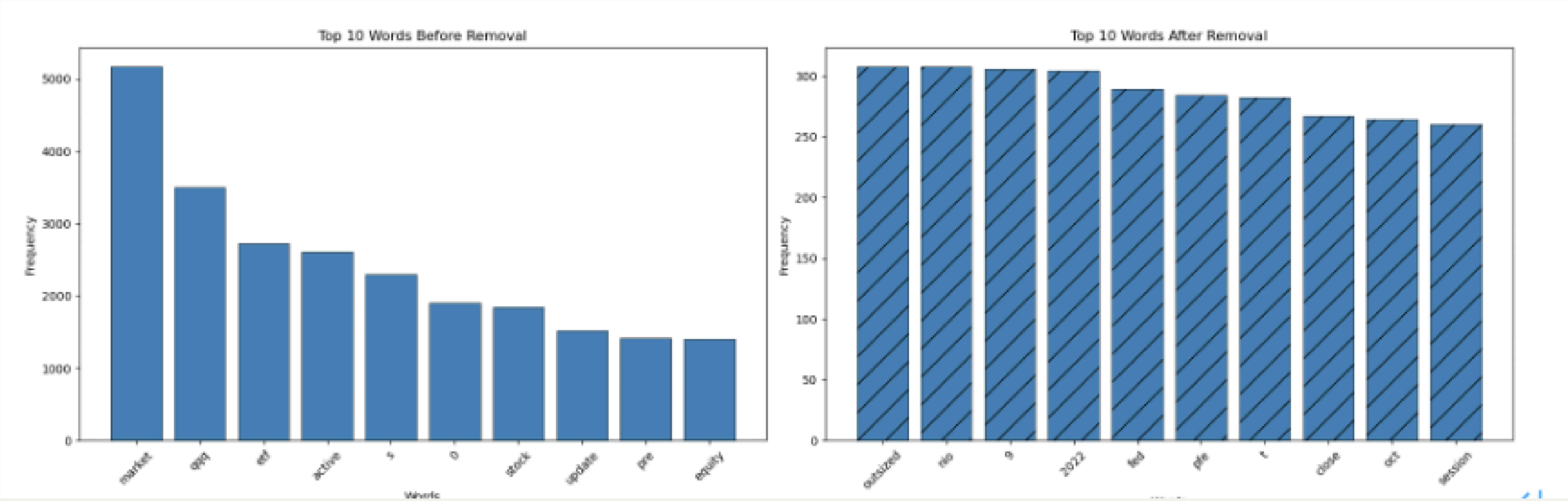
clean the noise, reduce bias

- "Word" to "word"
- Delete “a”, "and" or "the"
- "Running" simplified to "run."

High Freq Word Cleaning on News

clean the noise, reduce bias

- Excluded top 1% high frequency words



Before and After Cleaning

```
Sample Text Before and After High-Frequency Word Removal:
------------------------------------------------------------
Original: building permit , unemployment claim key thing watch week.
Filtered: building permit unemployment claim key thing .
------------------------------------------------------------
Original: 2 etf like invesco qqq trust 2024.
Filtered: like invesco trust 2024 .
------------------------------------------------------------
Original: guide nasdaq etf investing.
Filtered: guide investing .
------------------------------------------------------------
Original: pre - market profit conclude momentous trading week.
Filtered: profit conclude momentous .
------------------------------------------------------------
Original: invesco qqq etf turn millionaire.
Filtered: invesco turn millionaire .
------------------------------------------------------------
```

Text into sentimental score

use VADER, range from -1 to 1, it take the average of positivility of each word, so more positive word will have higher score

formula:

$$\text{StSc} = \frac{\sum(\text{Lexicon Score of Word} \times \text{Modifiers})}{\text{Number of Words}}$$

Text into sentimental score

use VADER, range from -1 to 1, it take the average of positivility of each word, so more positive word will have higher score

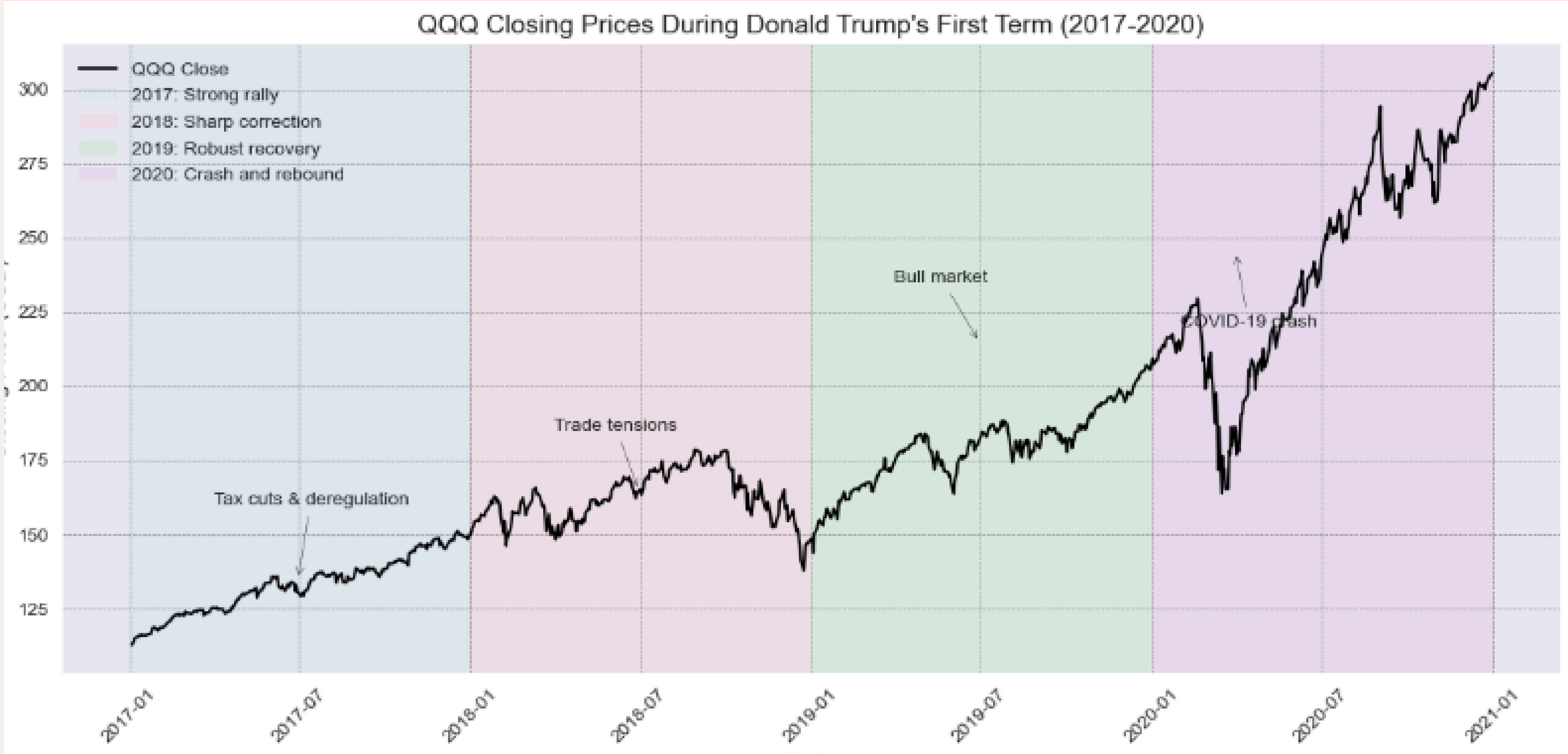
formula:

$$\mathrm{StSc} = \frac{\sum(\text{Lexicon Score of Word} \times \text{Modifiers})}{\text{Number of Words}}$$

Issue

- ~~No Target variable~~
- ~~News with noise, also need to transform into numeric data~~
- No prove that Trump 1st term can predict 2nd term
- No prove that DCA work
- lack of indicators
- Null value

Price trend of Trump's first term



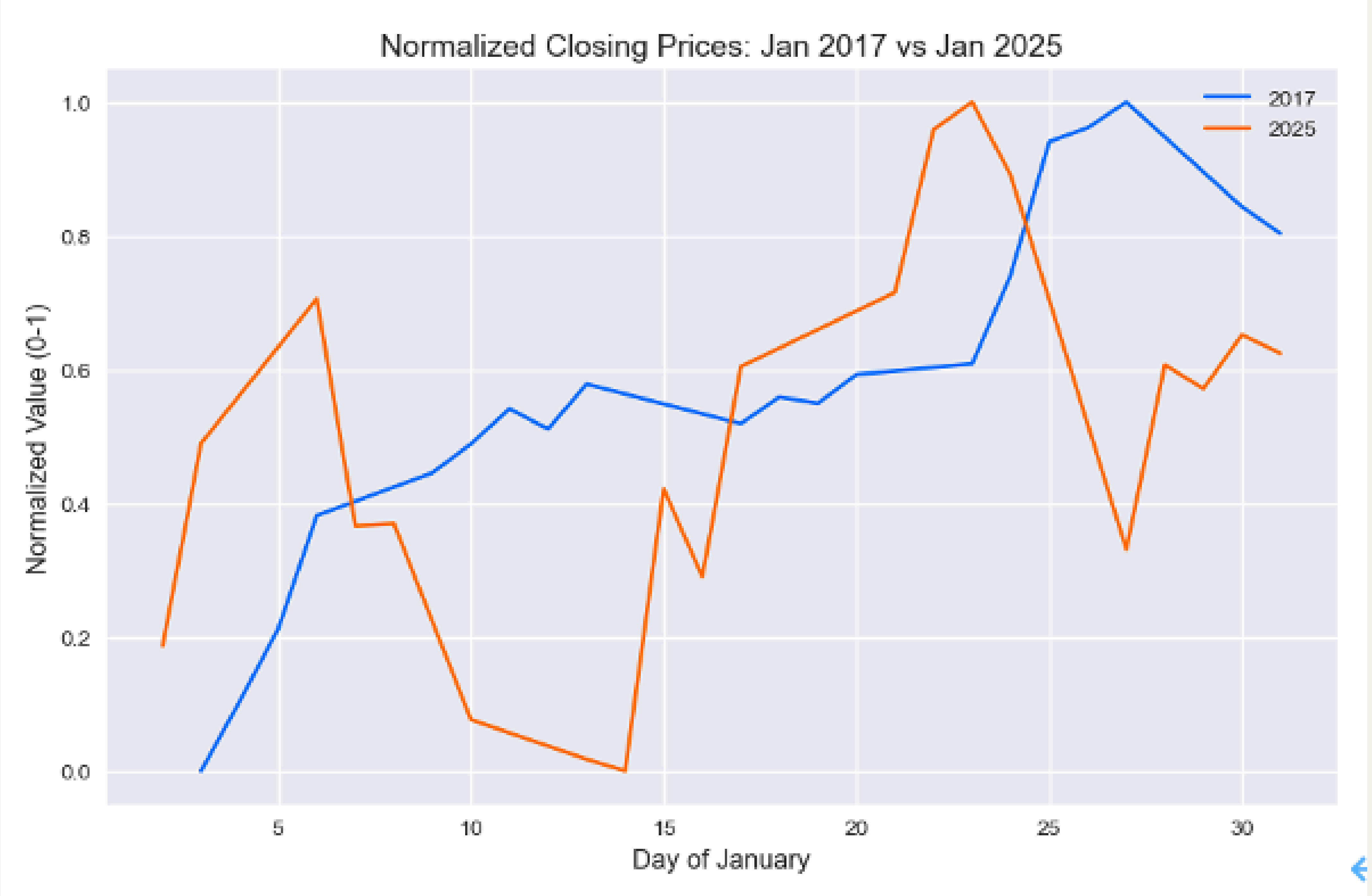
How to prove Trump 1st term can predict 2nd term?

- similarity in price trend and momentum (visalization)
- similarity in percentage change (pair t-test and corr)
- similarity in news titles (keywords, sentiment trend)
- similarity in economic situation (visualization)

Trump First Months in Office Similarity ('Trump Put' on Jan 2017 vs Jan 2025)

Normalized Closing Prices: Jan 2017 vs Jan 2025

-> Yes, both shown a increase trend.
-> Suggested that investor are excited about Trump being in office



Statistical similarity on Investment Behaviors

-> Despite volatility differences, a statistical test compared the average closing price trends in January 2017 and January 2025.

-> Pair t-test a p-value of 0.6678 (>0.05), there's no significant difference, confirming similar overall price trends in both periods.

-> In **finance**, the mean of **closing prices reflects the average trend**—whether it's increasing or decreasing.

```python
# Statistical similarity (paired t-test)
df_merged = pd.merge(
    data_2017[['Close_norm']].reset_index().assign(day=data_2017.index.day),
    data_2025[['Close_norm']].reset_index().assign(day=data_2025.index.day),
    on='day', how='inner'
)
stat, p_value = ttest_rel(df_merged['Close_norm_x'], df_merged['Close_norm_y'])
print(f"Paired t-test p-value: {p_value:.4f}")
print("Trends are similar" if p_value > 0.05 else "Trends are different")
```

```
Paired t-test p-value: 0.6678
Trends are similar
```

-> coef: is 0.27, showing that 27% of the pattern in 2017-01 is mirrored in 2025-01

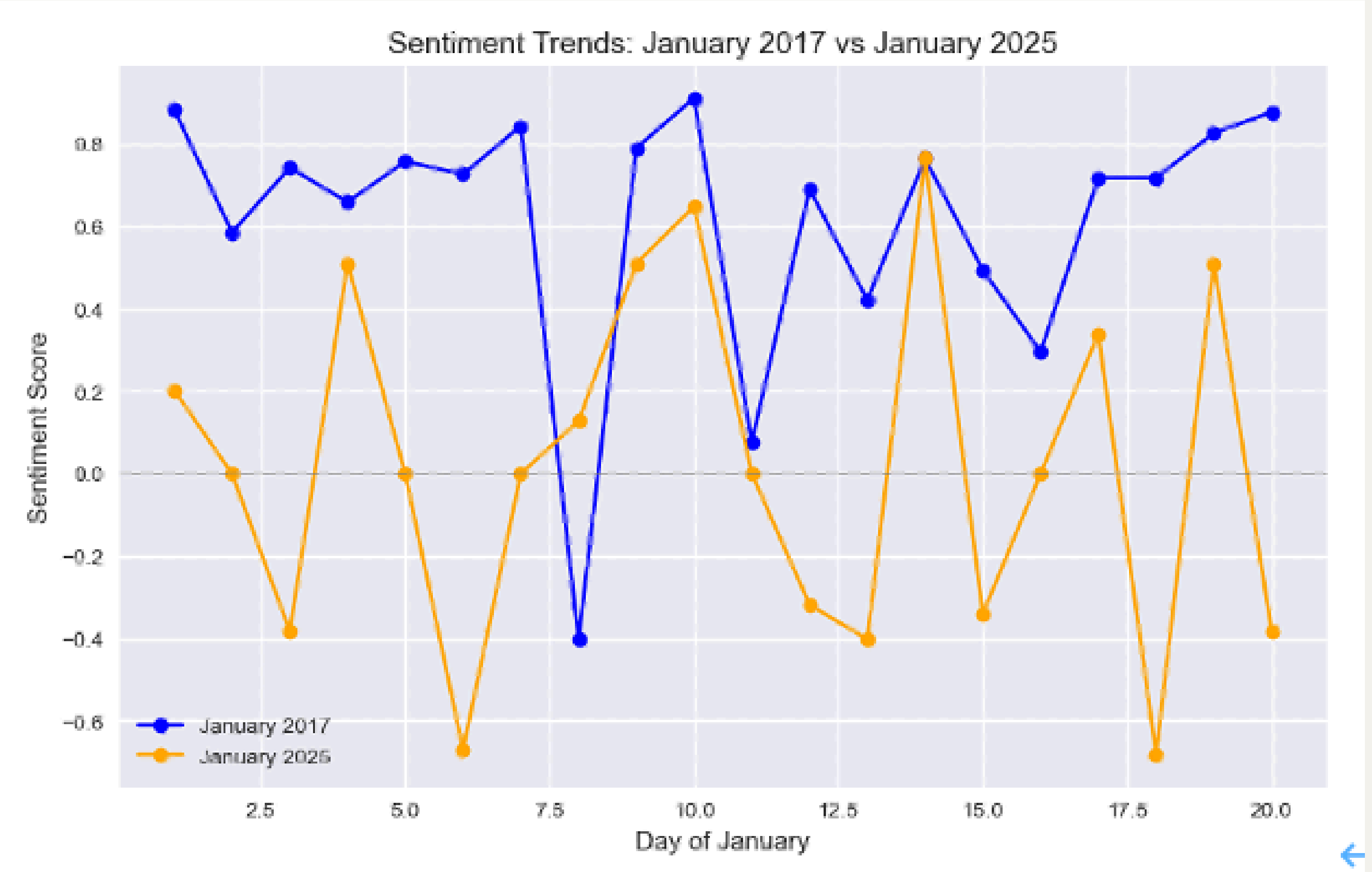
-> In stead of Trump excitement, There are other factor is difference on 2017-01, 2025-01

```python
# Correlation
correlation = np.corrcoef(data_2017['Close_norm'].dropna(), data_2025['Close_norm'].
print(f"Correlation between normalized closes: {correlation:.4f}")
```

```
Correlation between normalized closes: 0.2733
```

Similarity on Sentiment: January 2017 vs January 2025

Similarity on Sentiment: January 2017 vs January 2025



Statistical similarity on News sentiment

-> The proportion of 0.5789 indicates that sentiment changes in news articles aligned on approximately **57.89%** of the days.

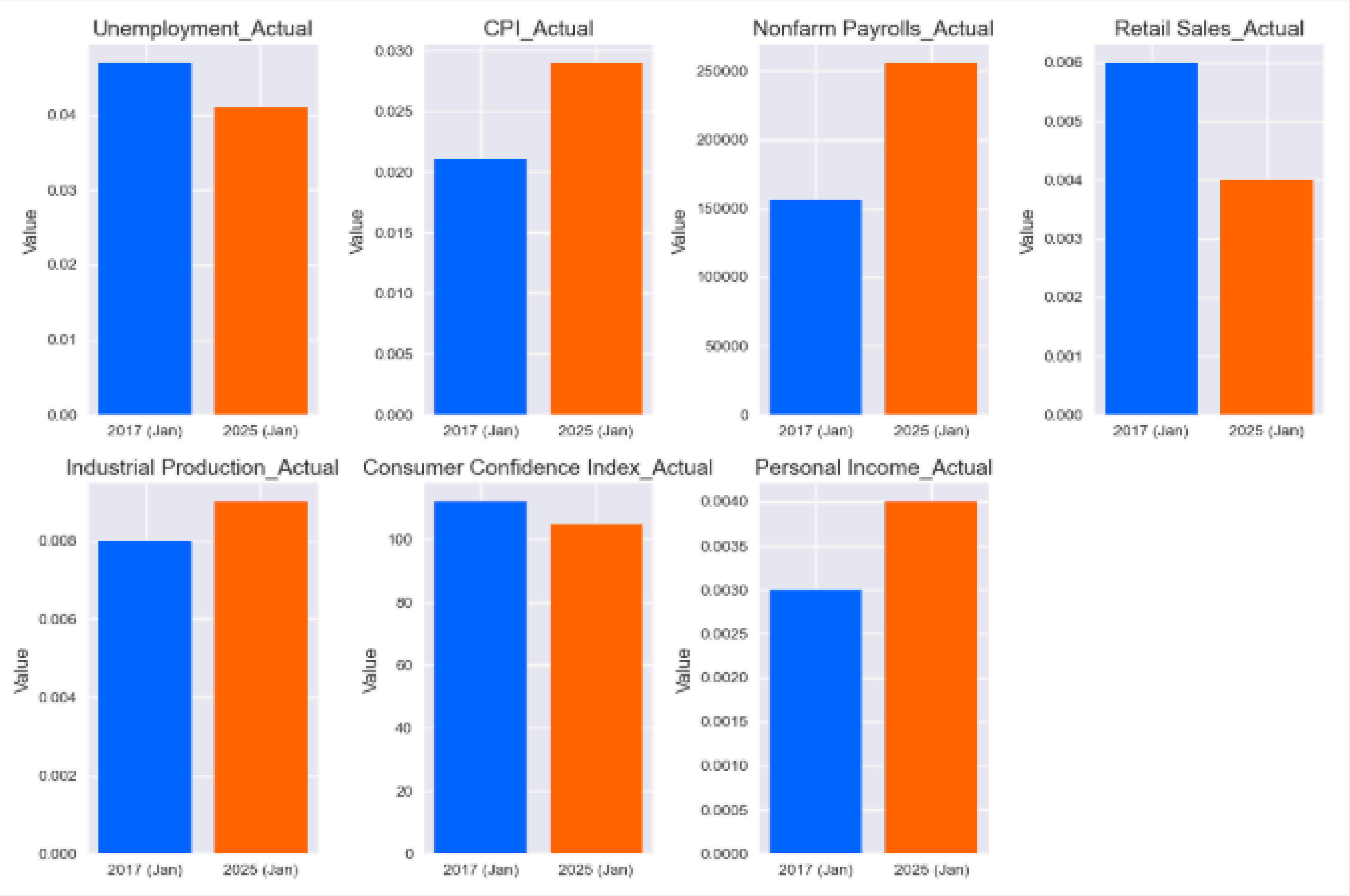
-> suggesting a moderate similarity in sentiment trends between January 2017 and January 2025.

```
# Calculate daily sentiment changes
data_2017['Sentiment_Change'] = data_2017['Sentiment'].diff()
data_2025['Sentiment_Change'] = data_2025['Sentiment'].diff()

# Determine the sign of sentiment changes and reset index
sign_2017 = np.sign(data_2017['Sentiment_Change'].dropna()).reset_index(drop=True)
sign_2025 = np.sign(data_2025['Sentiment_Change'].dropna()).reset_index(drop=True)

# Ensure both Series have the same length
min_length = min(len(sign_2017), len(sign_2025))
sign_2017 = sign_2017[:min_length]
sign_2025 = sign_2025[:min_length]

# Calculate the proportion of matching signs
matching_signs = (sign_2017 == sign_2025).mean()
print(f"Proportion of matching sentiment change directions: {matching_signs:.4f}")
```

```
Proportion of matching sentiment change directions: 0.5789
```

Macroeconomic Indicators: Jan 2017 vs Jan 2025

-> Mostly similar, or better off
-> Only difference is the Inflation is higher, and Retail Sales decrease



What did you know about the Trump Put?

- Price Information: Despite the violocity is more flucturate on 2025, Both of them show a increase trend, and the pair t-test with p-value of 0.6678, show that there is no significant difference between two trend.
- News: Keywords of trump and Pattern on sentiment shown similarity.
- Macro Attributes: Mostly similar, or better off. Only difference is the Inflation is higher, and Retail Sales decrease News: titles
- -> Trump's first month of presidency excited the market.

Trade War Similarity

-> Jul - Aug2018: The U.S., under President Trump, initiated a trade war by imposing tariffs on $34 billion worth of Chinese goods

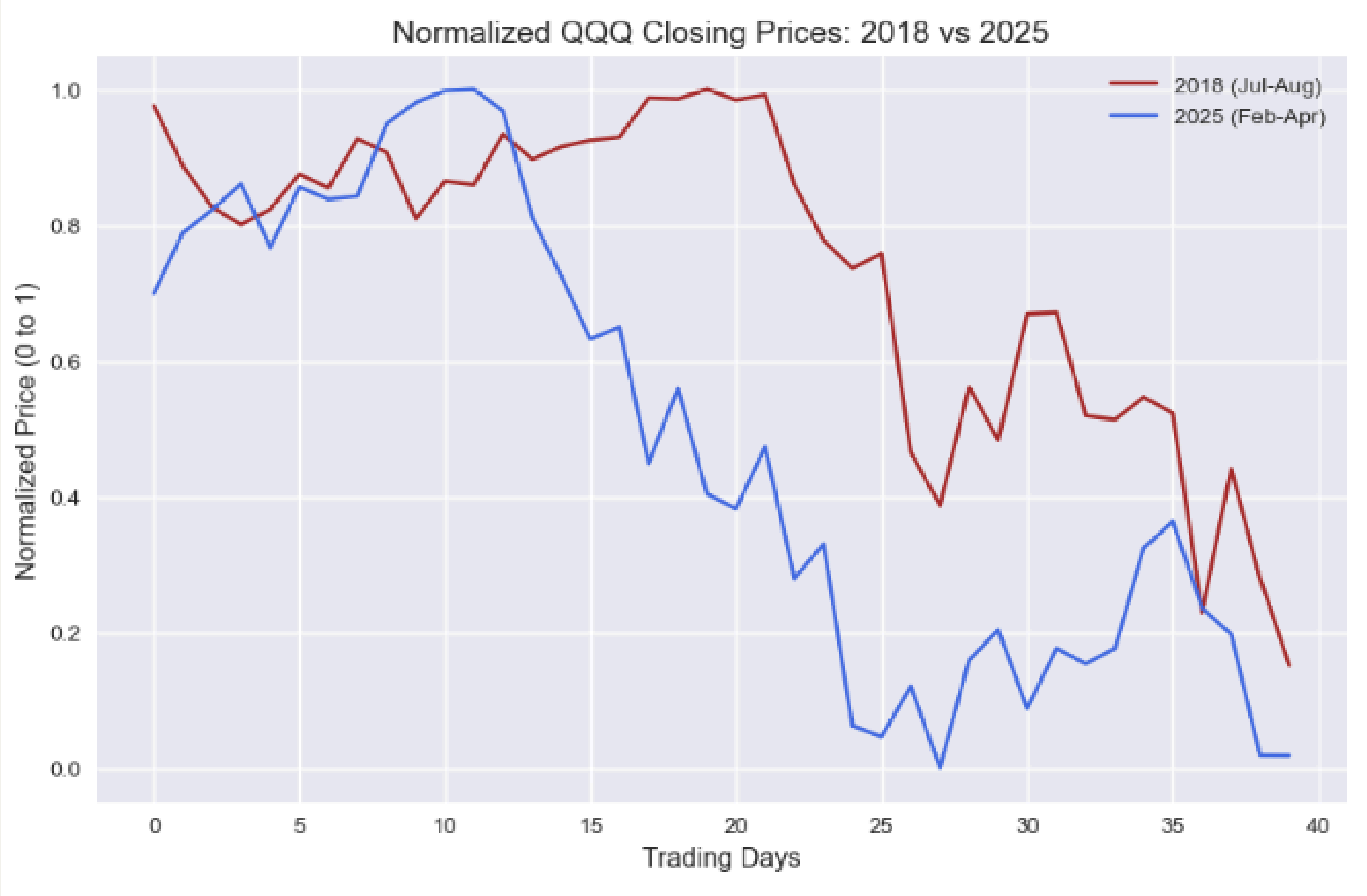
-> including cars and aircraft parts. China retaliated with tariffs on U.S. products like soybeans and automobiles, escalating tensions.

-> Feb-Apr 2025: Trump expressed views on using tariffs to "protect America,"

-> suggesting a continuation or escalation of trade policies, likely targeting China again, consistent with his earlier approach.

Normalized Closing Prices: Sep-Nov 2017 vs Feb-Apr 2025

-> Yes, both shown a decrease trend.
-> Suggested that investor are panic about Trade war



Statistical similarity on Investment Behaviors

-> Pair t-test a p-value of 0.6678 (>0.05)

-> there's no significant difference

-> confirming similar overall price trends in both periods.

-> In finance, the mean of closing prices reflects the average trend—whether it's increasing or decreasing.

```
t_stat, p_value = ttest_ind(tradewar_2018['Close_norm'].dropna(), tradewar_2025['Clo
print(f"Unpaired t-test statistic: {t_stat:.4f}, p-value: {p_value:.4f}")
if p_value > 0.05:
    print("The trends are similar (no significant difference).")
else:
    print("The trends are different (significant difference).")
```

```
Unpaired t-test statistic: 1.4117, p-value: 0.1621
The trends are similar (no significant difference).
```

-> coef: is 0.68

-> showing that 68% of the pattern in 2018 is mirrored in 2025

-> In stead of Trump excitement

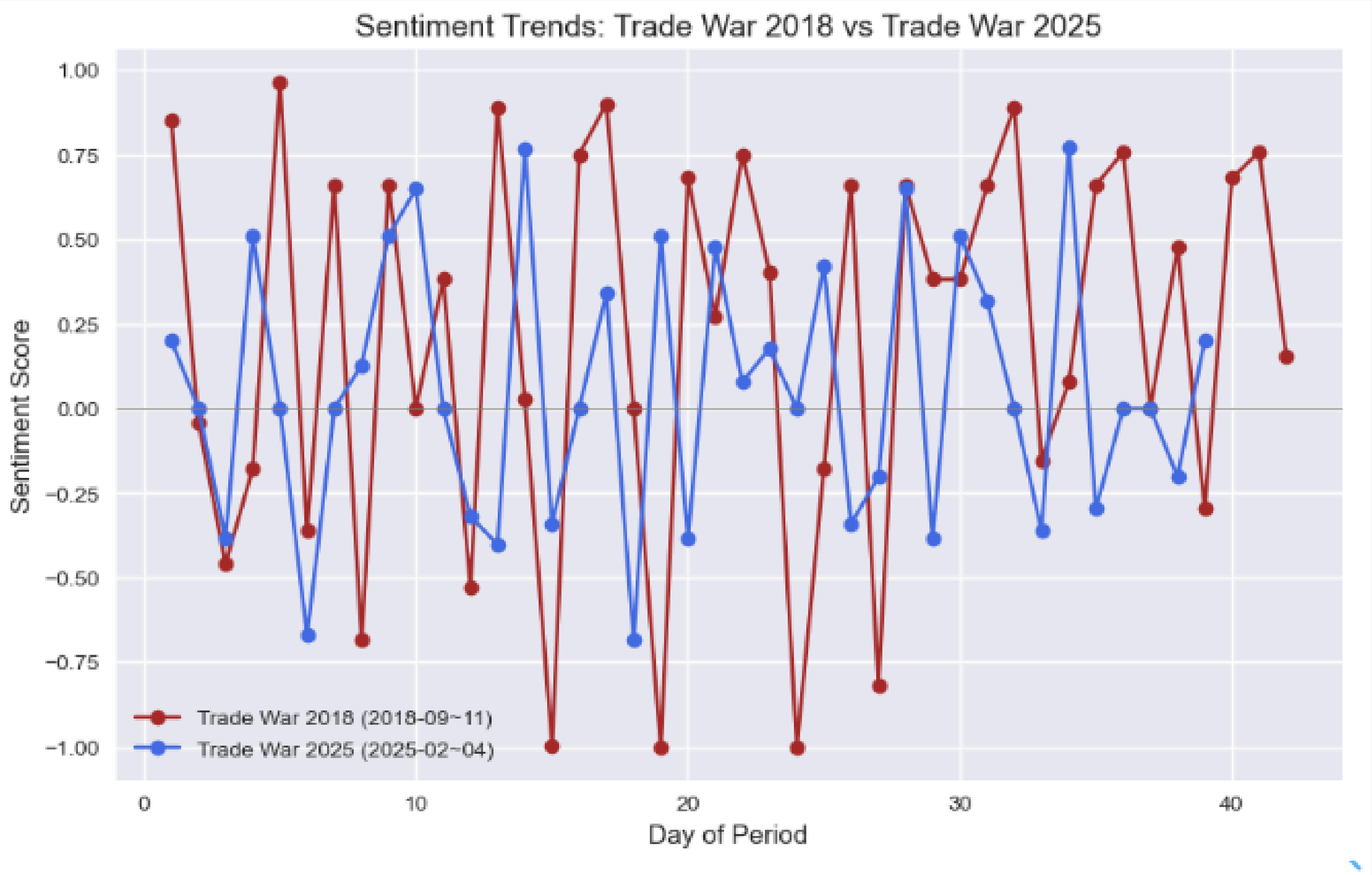
-> There are other factor is difference on 2018, 2025

```
# Your cleaning steps
data_2018_clean = tradewar_2018_trimmed['Close_norm'].dropna()
data_2025_clean = tradewar_2025_trimmed['Close_norm'].dropna()
min_length = min(len(data_2018_clean), len(data_2025_clean))
data_2018_clean = data_2018_clean[:min_length]
data_2025_clean = data_2025_clean[:min_length]

# Convert to NumPy arrays to avoid index issues
array_2018 = data_2018_clean.values  # Fixed: data_2017_clean -> data_2018_clean
array_2025 = data_2025_clean.values

# Calculate correlation using NumPy (bypasses pandas alignment)
correlation = np.corrcoef(array_2018, array_2025)[0, 1]
if np.isnan(correlation):
    print("Correlation is NaN. Check data integrity.")
else:
    print(f'Correlation between normalized closes: {correlation:.4f}')
```

```
Correlation between normalized closes: 0.6850
```

Similarity on Sentiment: Sep-Nov 2017 vs Feb-Apr 2025

Similarity on Sentiment: Sep-Nov 2017 vs Feb-Apr 2025



Statistical similarity on News sentiment

-> 47% of the sentimental change in new are similar of the day

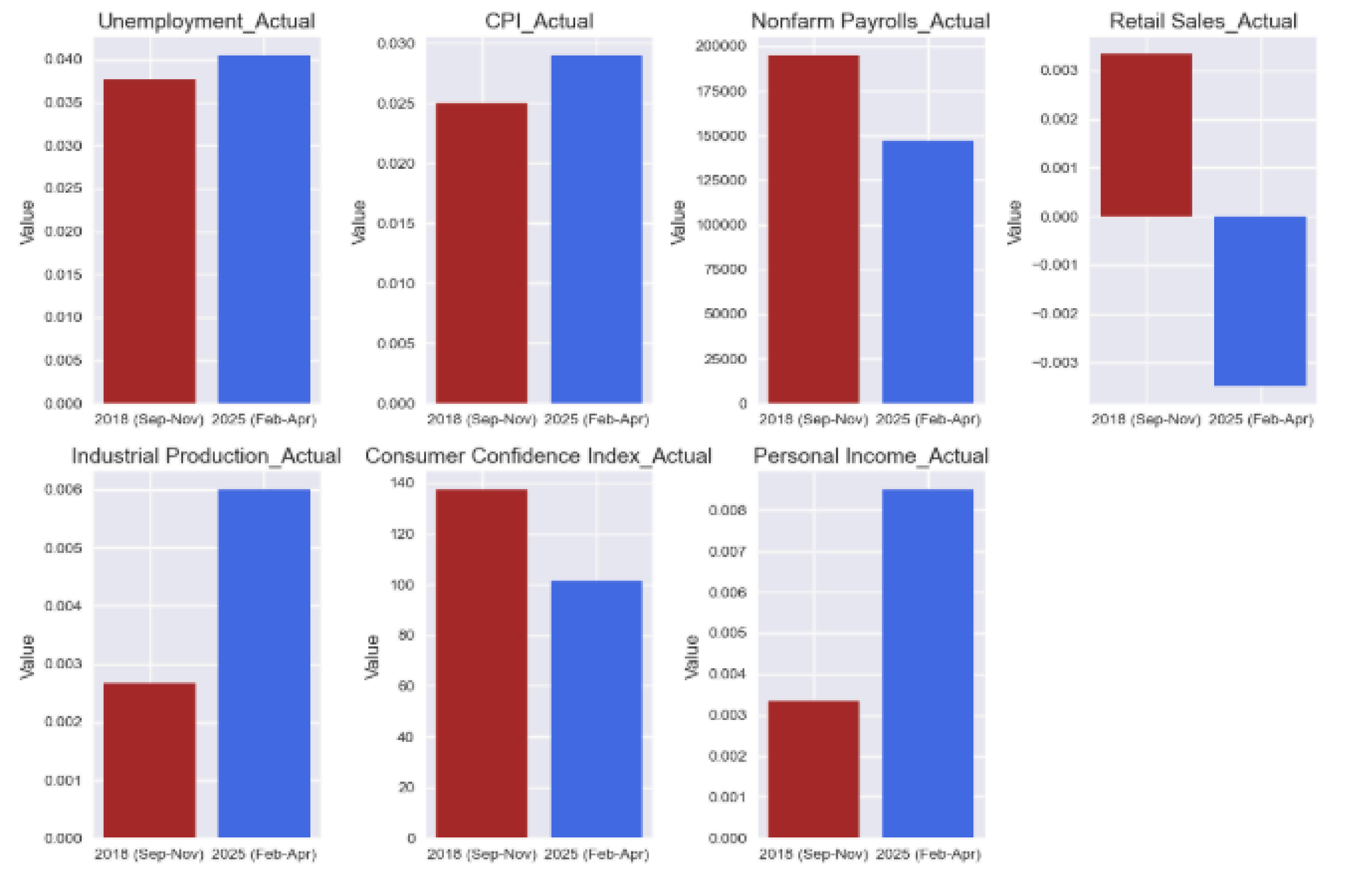
-> suggest a moderate similarity in sentimental trend

```
# Calculate the proportion of matching signs
matching_signs = (diff_2018 == sign_2025).mean()
print(f"Proportion of matching sentiment change directions: {matching_signs:.4f}")
```

```
Proportion of matching sentiment change directions: 0.4737
```

Macroeconomic Indicators: Sep-Nov 2017 vs Feb-Apr 2025

-> Unemployment, CPI, and Nonfarm Payrolls show similarity
-> suggesting a steady labor market and inflation despite trade tensions.
-> Industrial Production and Personal Income might improve (stronger activity and earnings)
-> while Retail Sales and Consumer Confidence could weaken (lower spending and sentiment), reflecting mixed economic impacts.



What did you know about the Trade War?

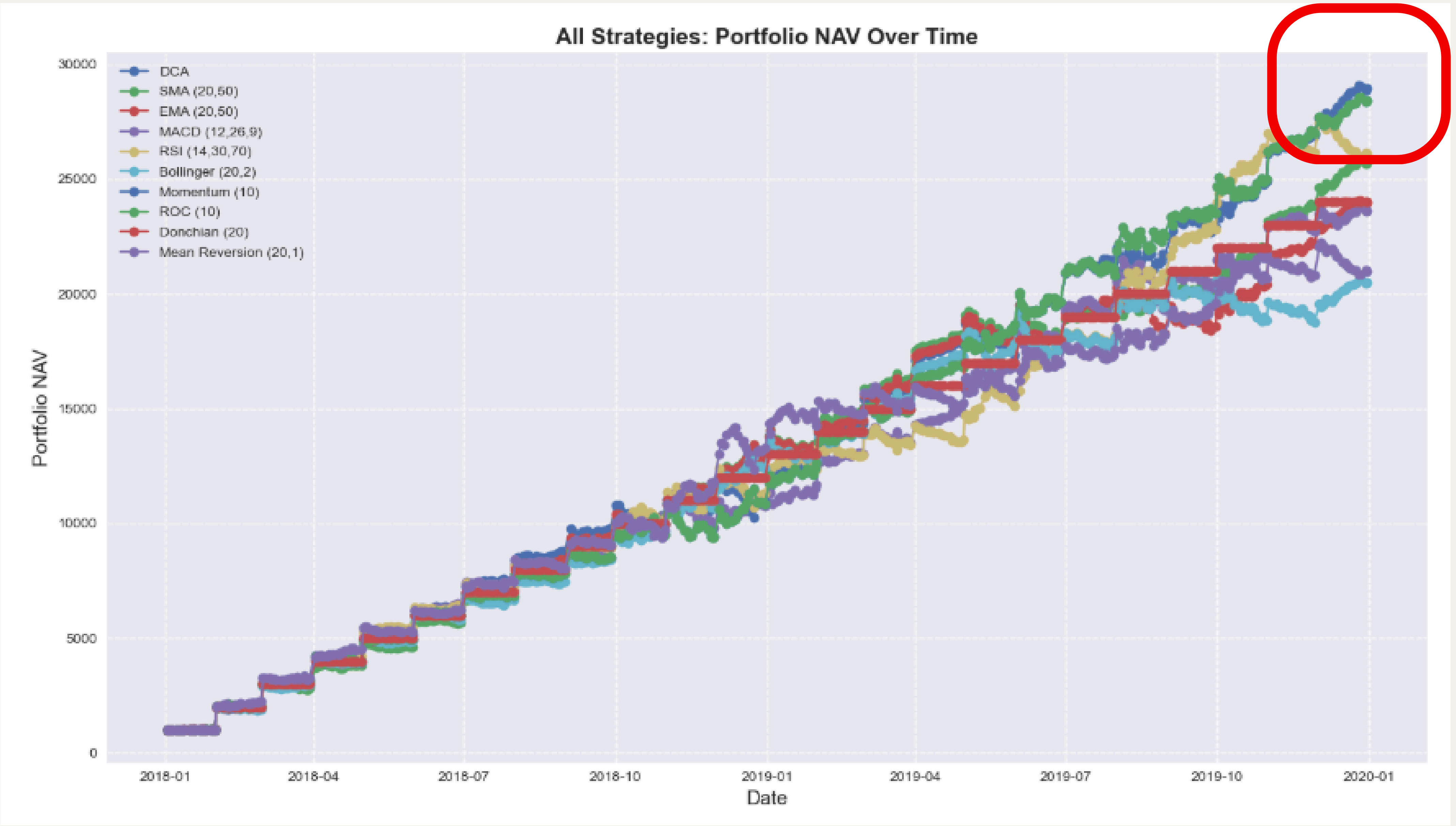
- Price Information: Both of them show a decrease trend, and the pair t-test with p-value of 0.1621, show that there is no significant difference between two trend.
- News: Keywords of "China" and "trade war", "Tariff" stand out, reflecting the U.S.-China tariff conflict. and Pattern on sentiment shown similarity.
- Macro Attributes: Unemployment, CPI, and Nonfarm Payrolls may show similarity, suggesting a steady labor market and inflation despite trade tensions. BUT, Industrial Production and Personal Income might improve (stronger activity and earnings), while Retail Sales and Consumer Confidence could weaken (lower spending and sentiment), reflecting mixed economic impacts.
- -> Trump’s approach to trade wars and tariffs in 2018 and 2025 exhibits similarities in market behavior, economic stability, and a focus on China.

Issue

- ~~No Target variable~~
- ~~News with noise, also need to transform into numeric data~~
- ~~No prove that Trump 1st term can predict 2nd term~~
- No prove that DCA work
- lack of indicators
- Null value

Backtest DCA on Trump 1st Presidency

- Dollar Cost Averaging (DCA): Invest a fixed amount (e.g., $1,000) at regular intervals, such as the last trading day of each month, regardless of price. Shares are accumulated over time, reducing the impact of volatility.
- SMA Crossover: Buy when a short Simple Moving Average (e.g., 50-day) crosses above a long SMA (e.g., 200-day). Sell or short when it crosses below.
- EMA Crossover: Same
- MACD: Buy when the MACD line (12, 26, 9 parameters) crosses above the signal line.
- Bollinger Bands Breakout: Buy when the price breaks above the upper band (e.g., 20-day SMA ± 2 standard deviations).
- Momentum: Buy when the price is above its n-day average (e.g., 10-day).
- Rate of Change (ROC): Buy when ROC (e.g., 10-day) is positive.
- Donchian Channel Breakout: Buy when the price exceeds the n-day high (e.g., 20-day).
- Mean Reversion: Buy when the price is significantly below its n-day mean (e.g., 20-day SMA minus 1 standard deviation).



Backtest Result

When comparing DCA with nine common strategies like SMA, Bollinger, and Mean Reversion, DCA gave the highest portfolio balance. profit of 21%.

Backtest Result

Strategy Performance Metrics:

	Sharpe ratio	CAGR	Win %	Avg. %	Avg. Win %	Avg. Loss %	Profit Factor	Max DD %	DD Duration	Recovery Factor
DCA	0.48	441.05%	56.37%	0.78%	1.97%	-0.82%	3.41	-15.07%	34 days	185.4
SMA (20,50)	0.46	409.7%	50.6%	0.76%	2.11%	-0.76%	3.51	-9.6%	58 days	257.31
EMA (20,50)	0.47	392.2%	55.38%	0.74%	1.93%	-0.73%	3.27	-9.96%	85 days	230.52
MACD (12,26,9)	0.47	388.85%	52.99%	0.74%	2.13%	-0.82%	2.93	-14.73%	57 days	153.67
RSI (14,30,70)	0.46	413.81%	48.61%	0.77%	2.27%	-0.69%	3.27	-13.83%	31 days	181.52
Bollinger (20,2)	0.43	355.14%	47.21%	0.72%	2.3%	-0.75%	2.99	-10.2%	119 days	191.22
Momentum (10)	0.48	436.15%	54.58%	0.78%	2.1%	-0.84%	3.13	-11.43%	29 days	239.82
ROC (10)	0.48	436.15%	54.58%	0.78%	2.1%	-0.84%	3.13	-11.43%	29 days	239.82
Donchian (20)	0.46	392.58%	4.58%	0.74%	16.24%	nan%	NaN	0.0%	0 days	NaN
Mean Reversion (20,1)	0.45	360.33%	48.21%	0.72%	2.28%	-0.8%	2.88	-12.75%	53 days	156.68

To look closer, DCA has a Sharpe ratio of 0.48, which means it gives a decent return for the risk, balancing profit and safety.

Issue

- ~~No Target variable~~
- ~~News with noise, also need to transform into numeric data~~
- ~~No prove that Trump 1st term can predict 2nd term~~
- ~~No prove that DCA work~~
- lack of indicators
- Null value

Why lack of indicators?

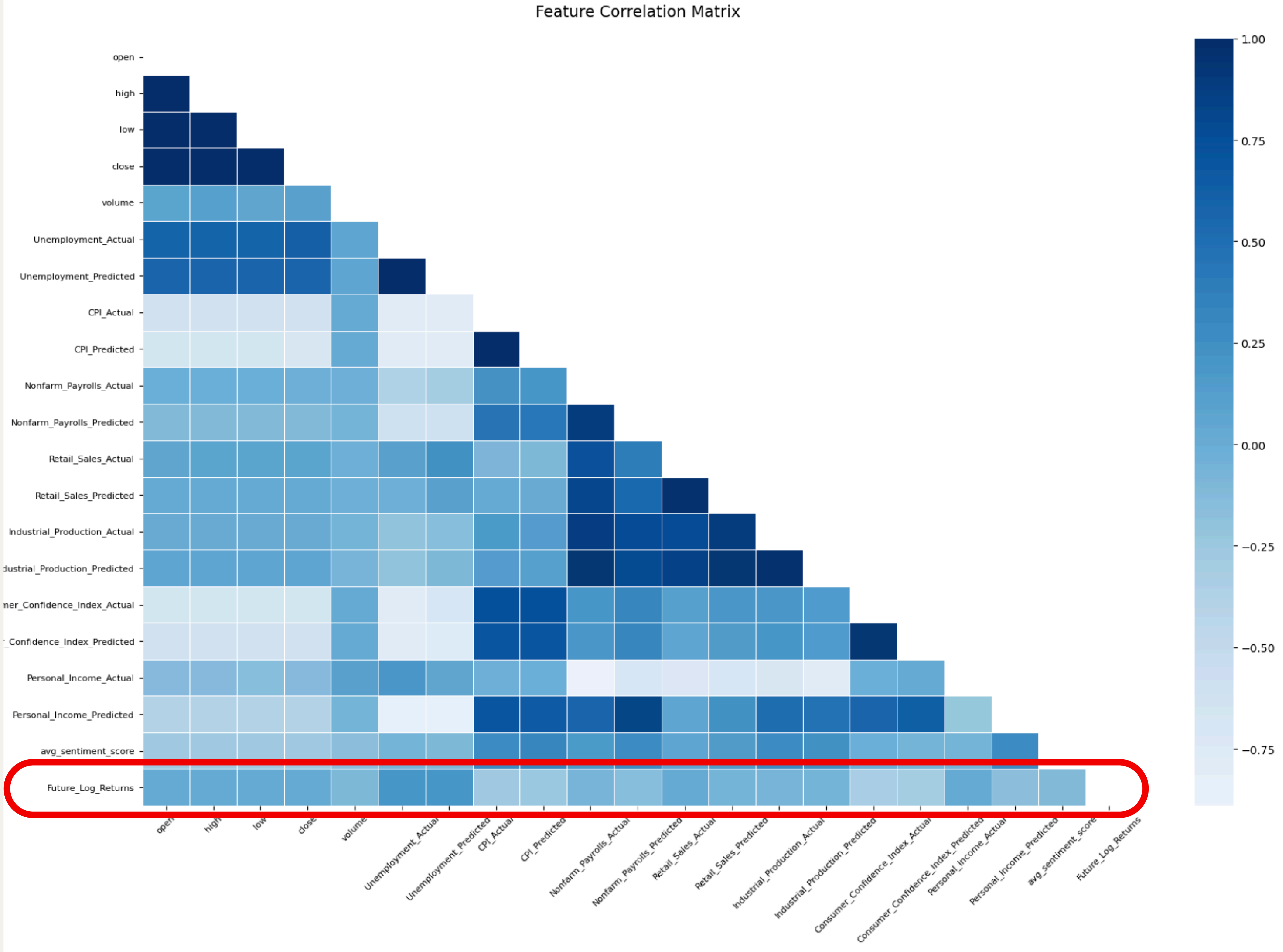
- lack of feature that correlated with target

correlation check

Correlation is find the relationship between two random variables.

Fit the assumption of find the most relevant for the model, reducing noise and improving model performance by excluding irrelevant or redundant features.

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$



Feature Correlation Matrix
open
high
low
close
volume
Unemployment_Actual
Unemployment_Predicted
CPI_Actual
CPI_Predicted
Nonfarm_Payrolls_Actual
Nonfarm_Payrolls_Predicted
Retail_Sales_Actual
Retail_Sales_Predicted
Industrial_Production_Actual
Industrial_Production_Predicted
Consumer_Confidence_Index_Actual
Consumer_Confidence_Index_Predicted
Personal_Income_Actual
Personal_Income_Predicted
avg_sentiment_score
Future_Log_Returns
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

Feature Construction - competitors

- Financial indicators like the VIX Index, Bitcoin, HangSeng Index, bond, gold and other 16 prices provide external context to stock movements.
- fetch these from Yahoo Finance.
- These indicators together offer a global perspective, helping the model consider macroeconomic conditions and sector-specific dynamics that influence stock prices.

Feature Construction – Technical Indicator

- High, Low price is not enough
- Technical indicators like SMA, MACD, RSI, Standard deviation can spotting patterns and trends within the stock market.
- fetch these from Pandas TA.
- These indicators together capture trend and momentum.

Feature Construction – Macro Feature

- Actual and Forcast macro data is not enough
- Expectation is important in finance, actual lower than expectation will disappoint investor, potentially causing price drop, so I calculate the difference on actual and focast CPI, Ump rate...
- To capture change in macro data, I add CPI change, Ump rate change by minus current months CPI.

Feature Construction - Lagged Feature

- In time series, past values inform the future. We add lagged features
- Lagged features like close_lag10 capture temporal dependencies
- I add lags=[1, 10, 42, 63] to capture short term and long term dependencies.

Too many features

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2333 entries, 0 to 2332
Columns: 295 entries, date to ^BVSP_lag63
dtypes: datetime64[ns](1), float64(288), int64(2), object(4)
memory usage: 5.3+ MB
None
```

Feature Selection: Recursive Feature Elimination with Cross-Validation (RFECV)

- The RFECV process begins by training a tree-based model, like a decision tree or random forest, using all features in the dataset.
- It calculates each feature's importance, base on Gini importance.
- higher score means the feature is more related to the target.
- To make sure it correct, cross-validation, it iteratively removes the least important features one by one, retraining the model each time, until it reaches the perfect features.

Selected 50 features:

	feature	importance
72	^VXN_lag10	0.095380
76	000001.SS_lag42	0.079458
36	Nonfarm_error	0.061208
82	^N225_lag42	0.044207
4	Nonfarm_Payrolls_Actual	0.035530
48	^VIX_lag10	0.028841
90	^MXX_lag10	0.028659
46	volume_lag42	0.023146
63	VXUS_lag1	0.022174
80	^N225_lag1	0.022036
25	MACD	0.021669
53	BZ=F_lag42	0.019653
28	RSI_14	0.018533
89	^MXX_lag1	0.018483
21	^FTSE	0.016503
52	BZ=F_lag10	0.015021
98	^FCHI_lag63	0.014634
109	^BVSP_lag42	0.014062
11	Personal_Income_Actual	0.013693
54	BZ=F_lag63	0.013053
44	volume_lag1	0.012638
34	OBV	0.012534
49	^VIX_lag42	0.011772
71	^VXN_lag1	0.011406
45	volume_lag10	0.011378
85	^GDAXI_lag63	0.011246
12	Personal_Income_Predicted	0.010465
1	volume	0.009606
8	Industrial_Production_Predicted	0.009176
19	^GDAXI	0.008733
20	^MXX	0.008445
51	BZ=F_lag1	0.008433
47	volume_lag63	0.008353
83	^GDAXI_lag10	0.008042
102	^HSI_lag63	0.007805
81	^N225_lag10	0.007721
87	^GSPTSE_lag42	0.007543
59	BTC-USD_lag63	0.007507
101	^HSI_lag42	0.007071
62	IXN_lag63	0.006612
96	^FCHI_lag1	0.006488
73	^VXN_lag42	0.006411
56	BTC-USD_lag1	0.006373
69	^GSPC_lag10	0.006241
6	Retail_Sales_Predicted	0.005855
24	EMA_50	0.005542
95	^FTSE_lag63	0.005542
75	000001.SS_lag1	0.005521
70	^GSPC_lag63	0.005114
43	close_lag63	0.005021

Figure 7: Finalize 50 feature selected

Feature Selection Result

Total 50 feature selected

^VXN_lag10 (the price of reverse fund of QQQ 10 days ago) holds an importance score of 9%, rank first place.

000001.SS_lag42 (the price of china index 42 days ago) has an importance score of 8%, rank 2, rank second place.

To make it New feature normal distributed, Log transformation is used

all feature take log, the value now between 0 to 1., which avoid skew data

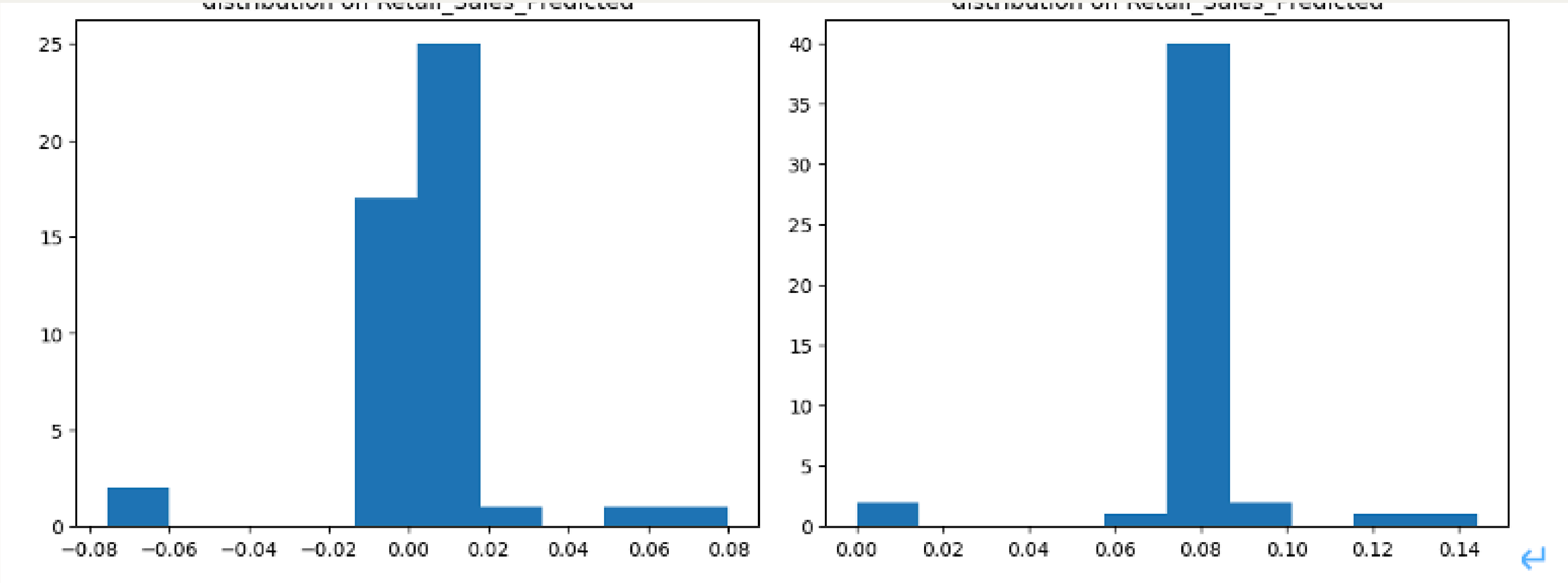


Figure 9: Before and after of log transformation on Retail_Sales_Predicted

To make it New feature in the same scale, Z-score Normalization is used

machine learning model will be mislead and weight some variable more important, to treat every variable equally, and avoid outliner all feature minus the mean and divide the standard deviation.

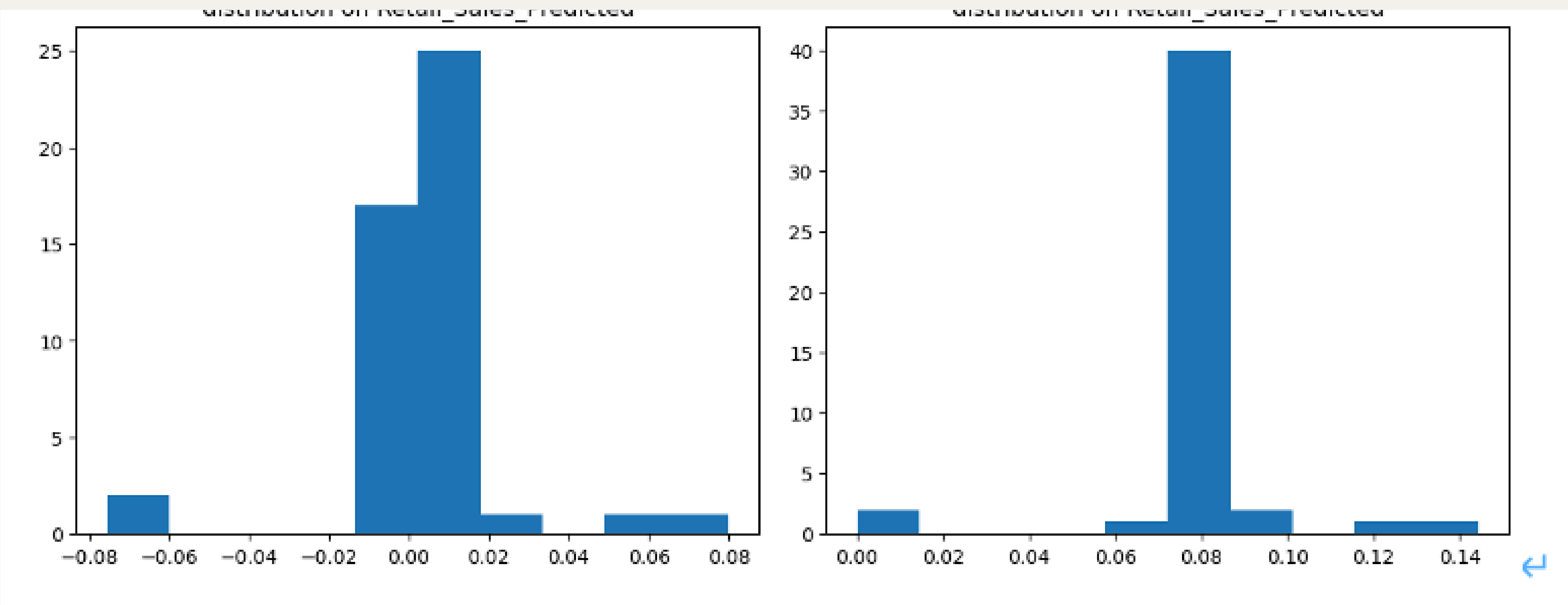


Figure 9: Before and after of log transformation on Retail Sales Predicted

Issue

- ~~No Target variable~~
- ~~News with noise, also need to transform into numeric data~~
- ~~No prove that Trump 1st term can predict 2nd term~~
- ~~No prove that DCA work~~
- ~~lack of indicators~~
- Null value

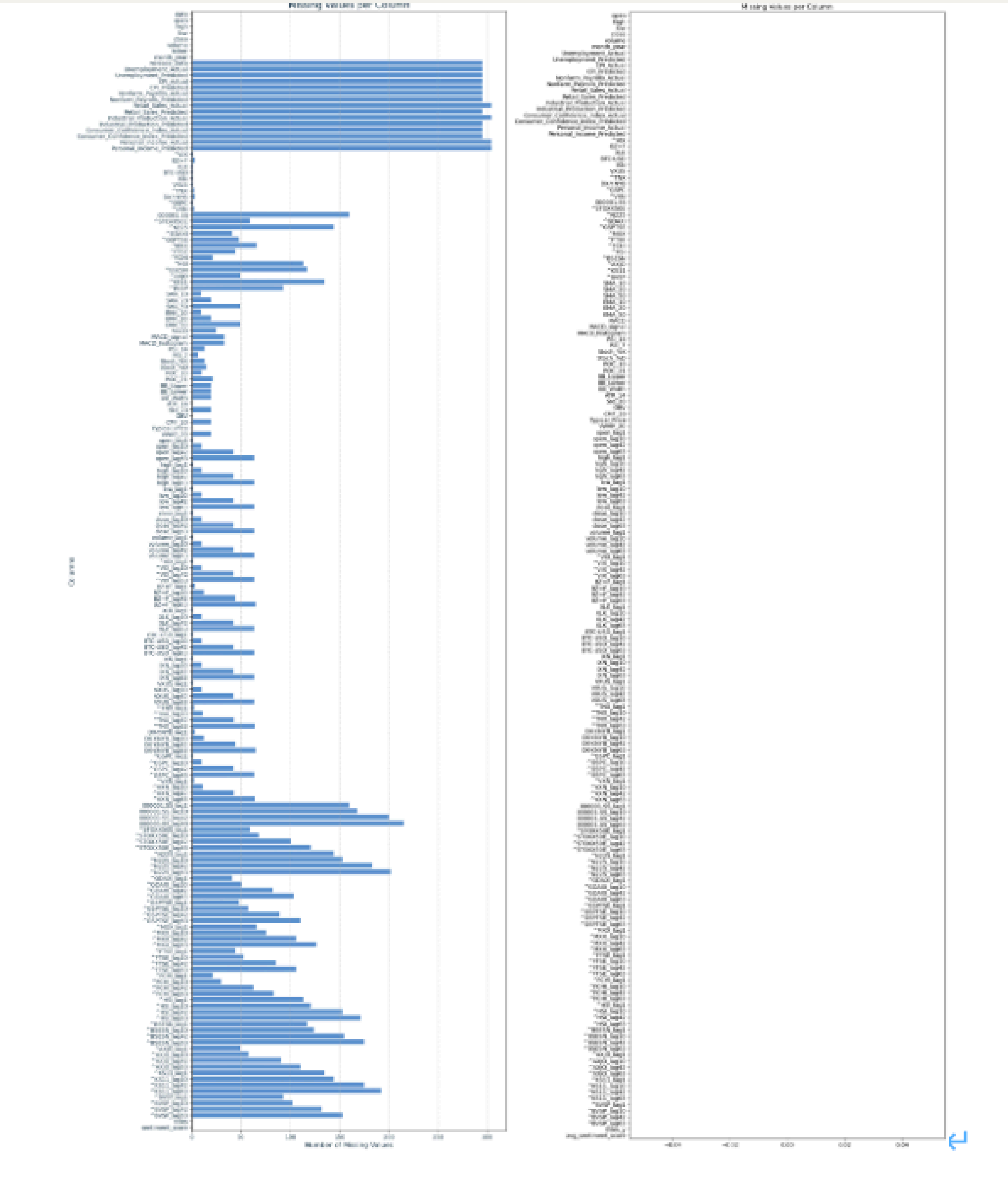


Figure 5: Before and after sorting date between 2017 and 2021

Remove Null Value

- Null value present because the lack of news data are on Biden period
- After include only only dates between 2017 and 2021
- There is no null value

Regression Model that applied

Linear Regression,
Vanilla LSTM,
Bidirectional LSTM,
CNN-LSTM

Linear regression

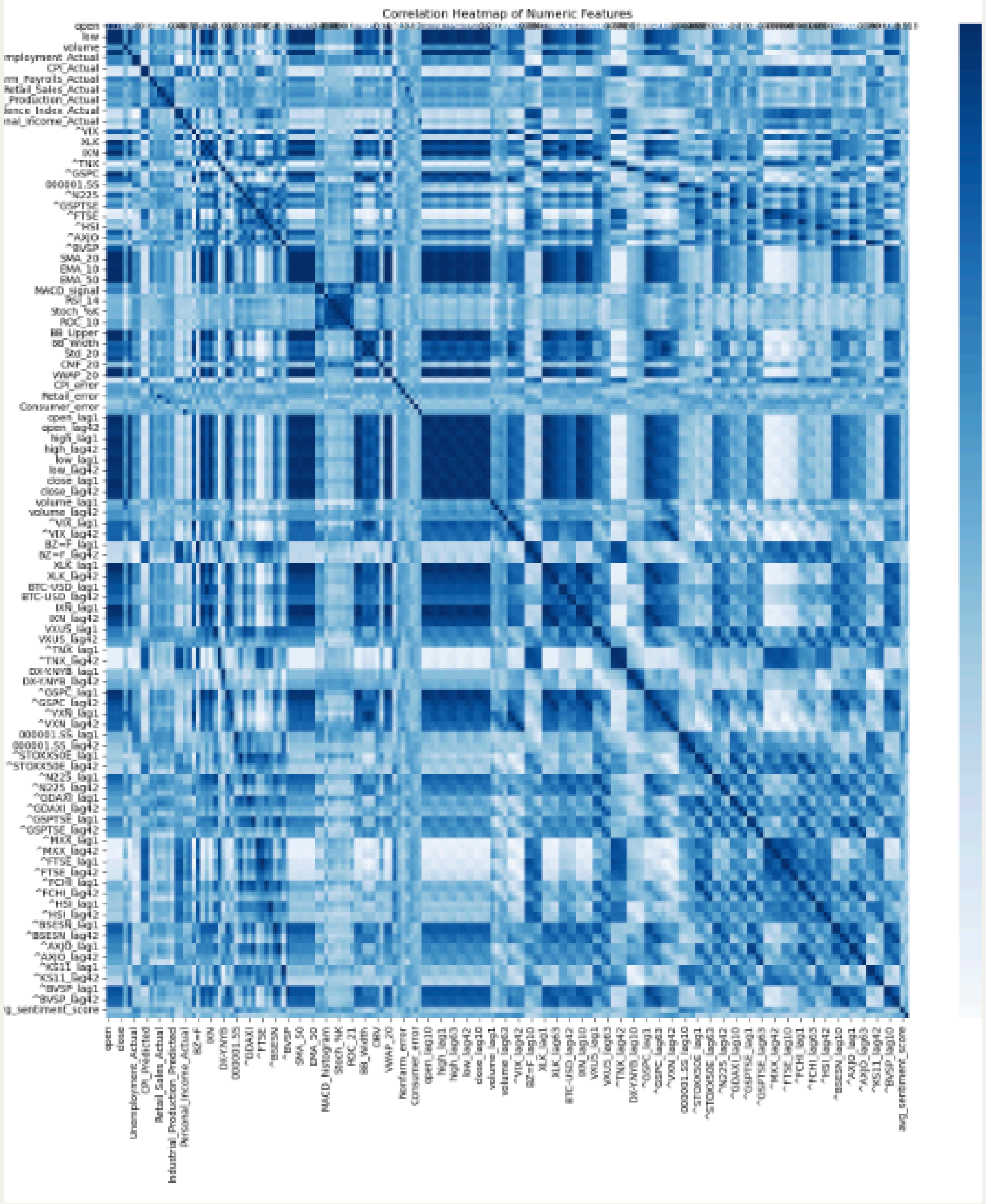
It capture the linear relation between feature and target “Future Log Return”

it calculate the intercept (baseline price holding other constant), and the coef between each variable and target.

and finally fitting a linear line to estimate the line

I treat it as the basedline model for comparison

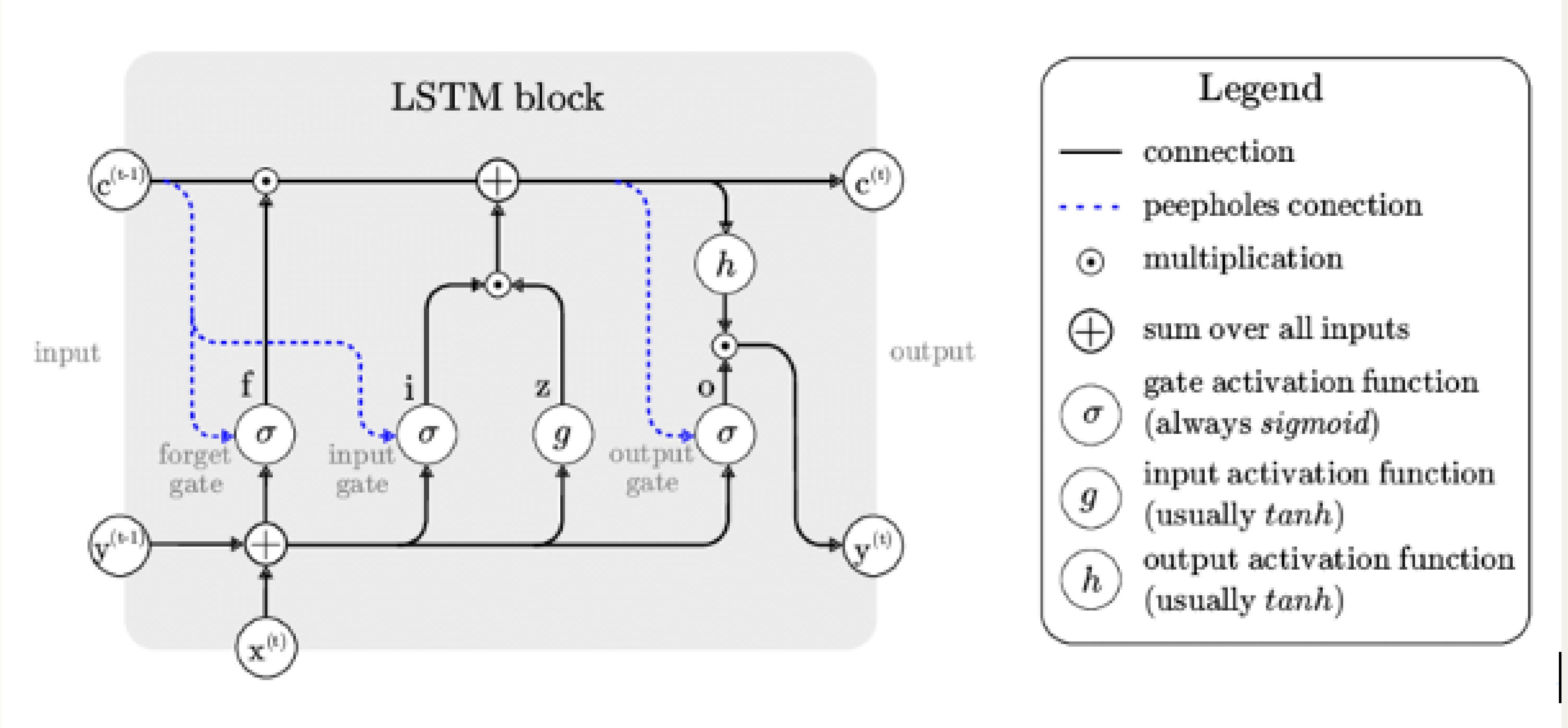
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ \dots \ + \beta_n x_n + \epsilon$$



Still, some feature is Non linear

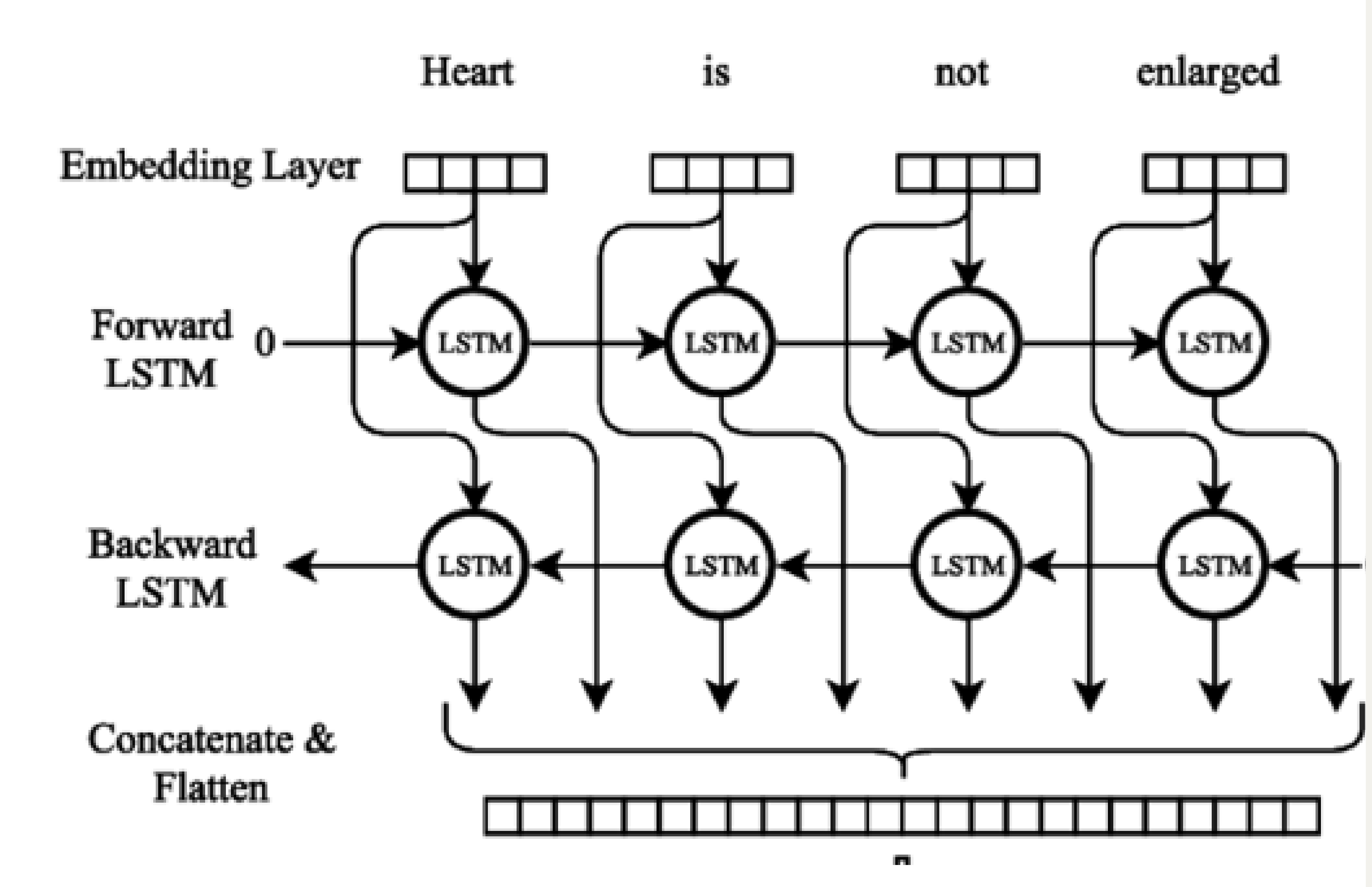
Dont fit the assumption of linear regression, so other model is employed

Vanilla LSTM



Long Short-Term Memory networks, are a type of recurrent neural network designed to model sequential data and capture temporal dependencies

The model have many memory cells that remember the information before, enabling the model to learn long-term dependencies. training forward to adjust weight to minimize error.



Bidirectional LSTM

Unlike Vanilla LSTM, it trains in both forward and backward directions

It can capture past and future points more accurately, understand the sequence as a whole, but may overfit, the line may become too squiggle.

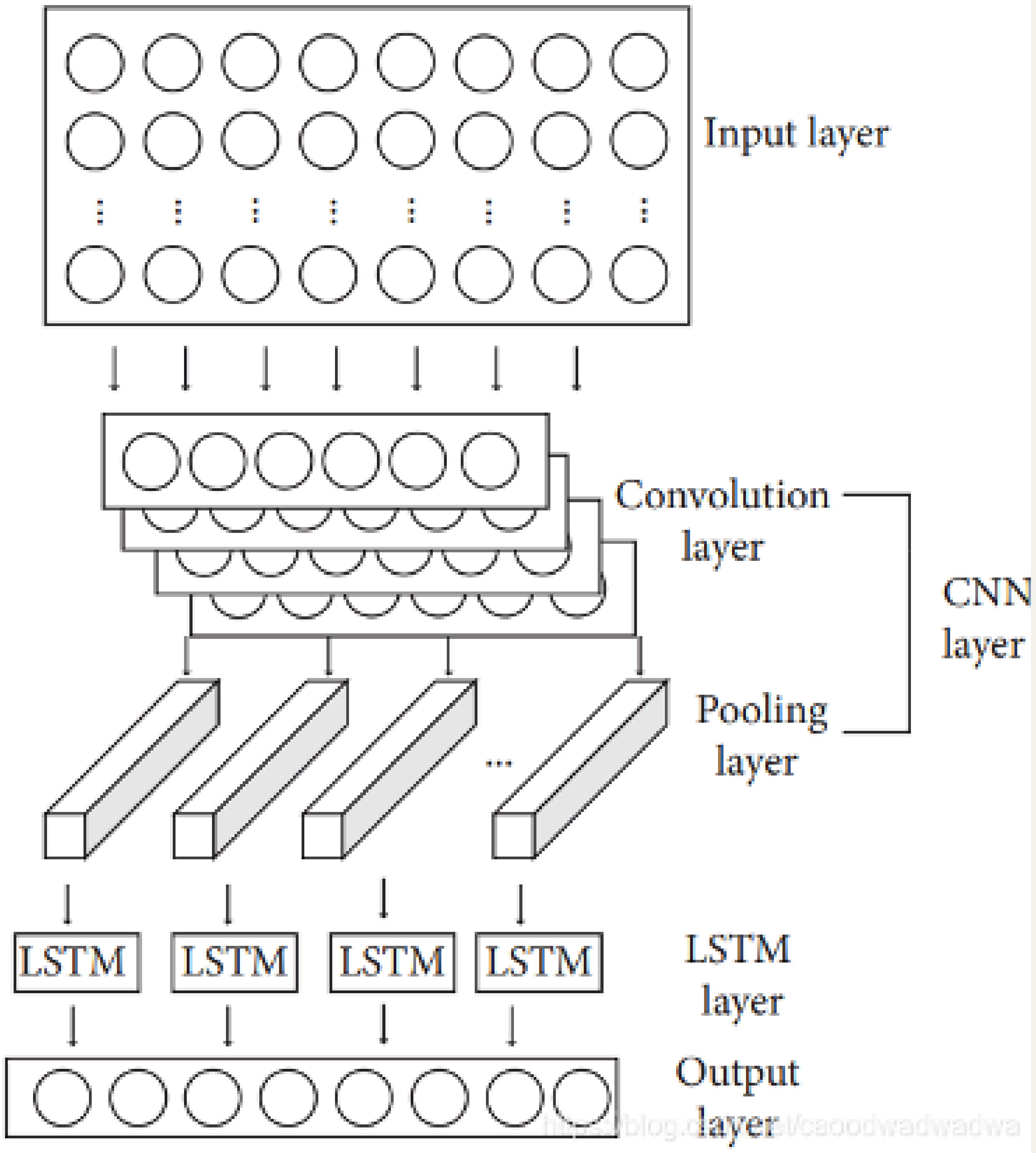


Image 13: Architecture of CNN-LSTM

CNN-LSTM

This hybrid approach take advantage on CNN and LSTM

first employs CNNs to extract data patterns and structures, and then uses LSTM networks to capture temporal dependencies.

Evalution

MSE

Cross validation

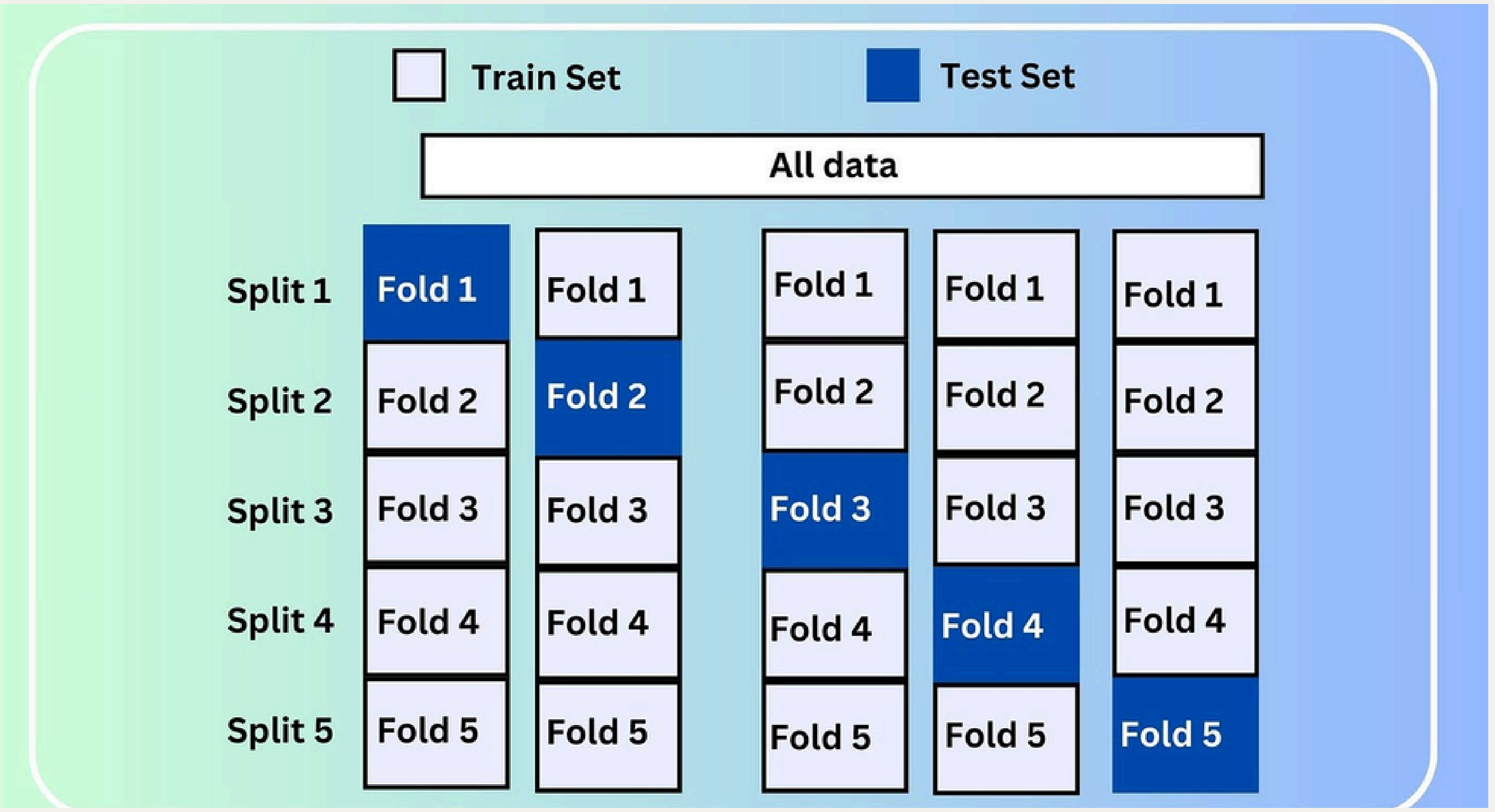
Mean Squared Error (MSE)

MSE can capture the average lose between testing and actual future log return.

calculates the average of the squared differences between predicted values and actual values.

provides an overall measure of a model's accuracy, and overfitting, with lower values indicating better performance.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$



5 fold Cross validation on MSE

One time maybe pure luck, because the fixed test set is fixed

to make sure the model preform well on unseen data, the dataset is divided into '5' subsets or folds, each time take different set for training, and the rest on test set.

and average the MSE

so it can ensure that the model's performance is consistent across different subsets of data

Result

Bidirectional LSTM achieve lowest MSE on 5 fold cross

	Linear Regression	Vanilla LSTM	Bidirectional LSTM	CNN-LSTM
MSE	0.098020	0.077488	0.066609	0.070871
RMSE	0.313082	0.278366	0.258087	0.266216
MAE	0.285050	0.265049	0.239464	0.249905

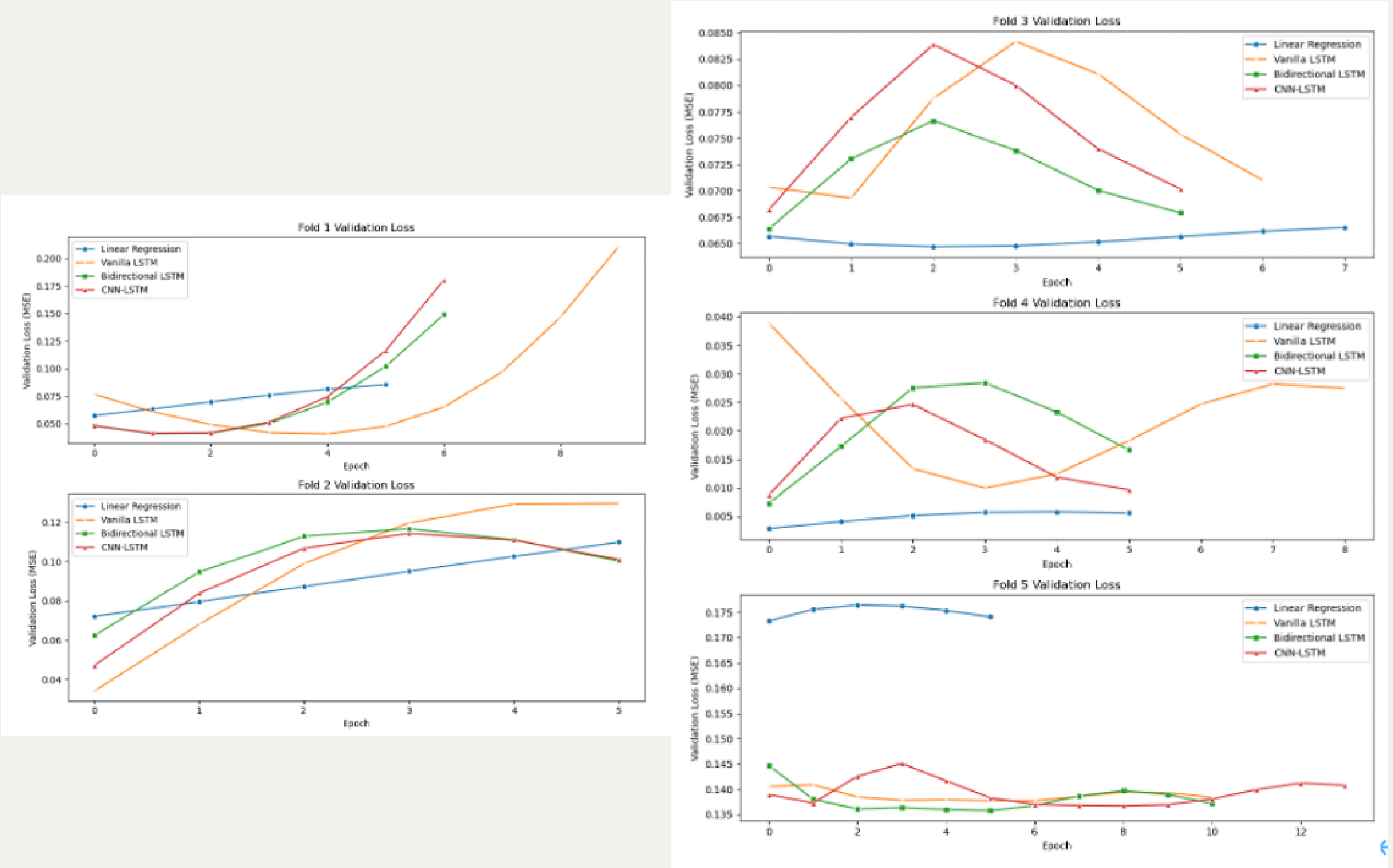


Image 23: 5 Fold Cross validation

To take a deeper look, Bidirectional LSTM (Green) is lower than others model in each fold

Fine-tuned Bidirectional LSTM

Model	Parameters
Bidirectional LSTM	Learning rates: 0.001, 0.0005, 0.0001 Hidden sizes: 50, 100, 150 Number of layers: 1, 2, 3 Dropout rates: 0.2, 0.3, 0.4 Batch sizes: 32, 64, 128 Activation functions: linear, Tanh, ReLU, Leaky ReLU, ELU

Table 2: Fine-Tuned Models Parameters

Trail and error on different combination and find the lowest average MSE in 5 fold cross validation

```
Testing: lr=0.001, hidden=150, layers=1, dropout=0.3, batch=128, activation=Re
LU()

Testing: lr=0.001, hidden=150, layers=1, dropout=0.3, batch=128, activation=Le
akyReLU(negative_slope=0.01)

Testing: lr=0.001, hidden=150, layers=1, dropout=0.3, batch=128, activation=EL
U(alpha=1.0)
Early stopping at epoch 17

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=32, activation=Non
e

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=32, activation=Tan
h()
Early stopping at epoch 33

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=32, activation=ReL
U()

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=32, activation=Lea
kyReLU(negative_slope=0.01)

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=32, activation=ELU
(alpha=1.0)
Early stopping at epoch 22

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=64, activation=Non
e

Testing: lr=0.001, hidden=150, layers=1, dropout=0.4, batch=64, activation=Tan
h()
Early stopping at epoch 44
```

```
BidirectionalLSTM(
  (lstm1): LSTM(50, 150, batch_first=True, bidirectional=True)
  (lstm2): LSTM(300, 150, batch_first=True, bidirectional=True)
  (norm): BatchNorm1d(300, eps=1e-05, momentum=0.1, affine=True, track_running
_stats=True)
  (dropout): Dropout(p=0.2, inplace=False)
  (fc1): Linear(in_features=300, out_features=150, bias=True)
  (fc2): Linear(in_features=150, out_features=1, bias=True)
  (activation): Identity()
)
```

```
Mean Squared Error (MSE): 0.0568
Root Mean Squared Error (RMSE): 0.2382
Mean Absolute Error (MAE): 0.2032
```

Best model:

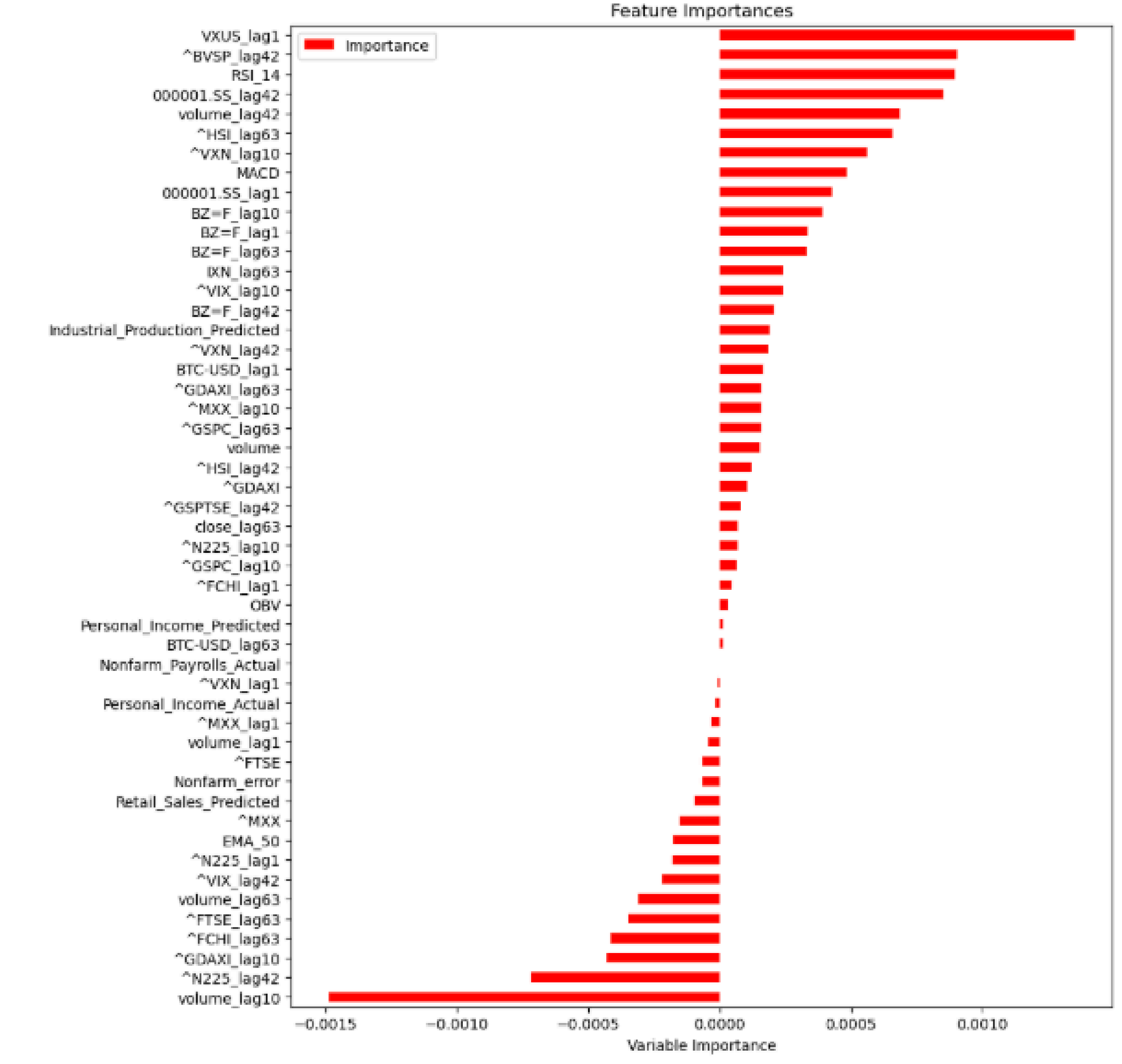
- **Learning Rate: 0.001.**
- **Hidden Size: 150 (300 output).**
- **Number of Layers: 2.**
- **Dropout Rate: 0.2.**
- **Batch Size: 64.**
- **Activation Function: Identity.**

Bidirectional LSTM
0.066609
0.258087
0.239464

```
Mean Squared Error (MSE): 0.0568
Root Mean Squared Error (RMSE): 0.2382
Mean Absolute Error (MAE): 0.2032
```

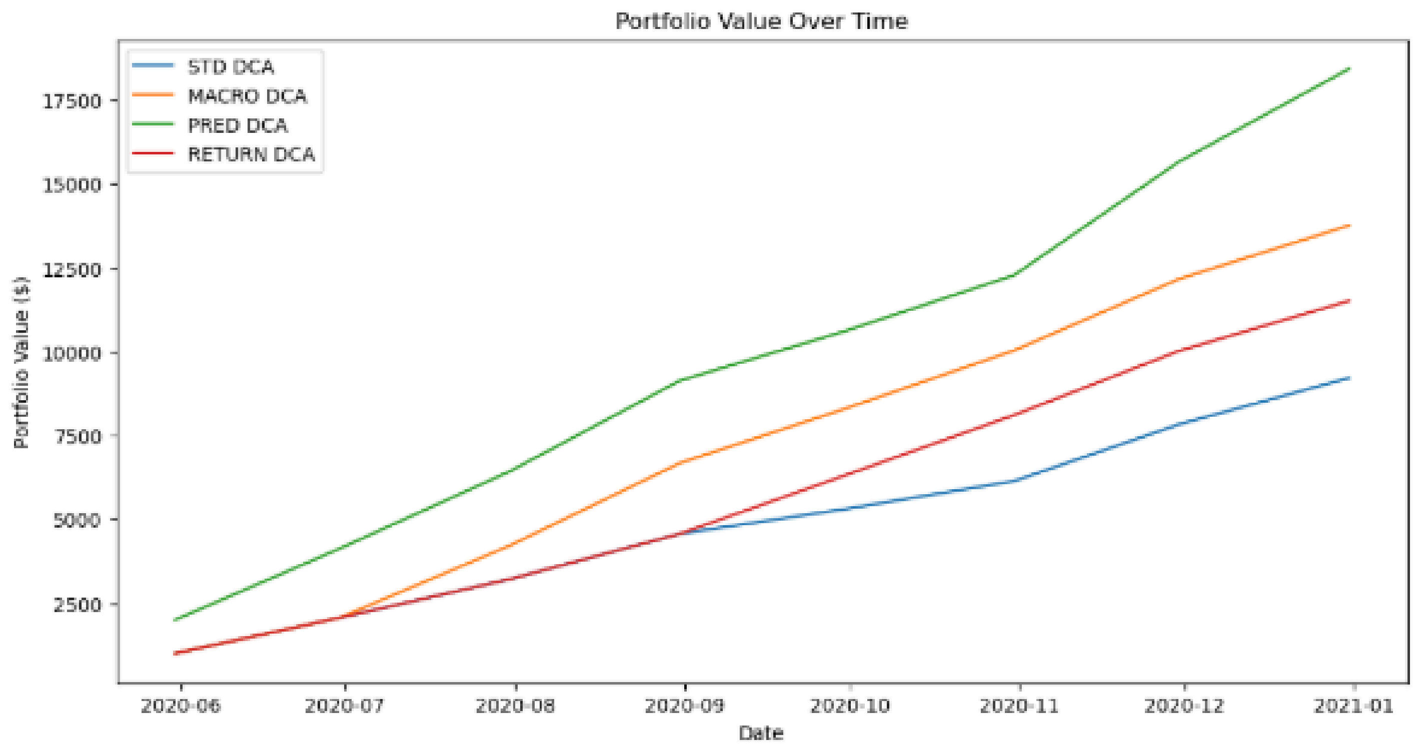
Best model:

- MSE was reduced by 2%,
- predictions were closer to the real data, prove that the model’s is able to make good predictions on new data was improved as a result.



The price of reverse QQQ of 1 day ago and QQQ's volume of 10 day before have largely affect the future log return of next month

```
Backtest Metrics for Each Strategy (Last 8 Periods):

STD DCA
Total Return (%): 15.18
Annualized Return (%): 15.18
Sharpe Ratio: 3.74
Volatility (%): 32.81

MACRO DCA
Total Return (%): 14.67
Annualized Return (%): 14.67
Sharpe Ratio: 3.75
Volatility (%): 40.58

PRED DCA
Total Return (%): 15.18
Annualized Return (%): 15.18
Sharpe Ratio: 3.74
Volatility (%): 32.81

RETURN DCA
Total Return (%): 15.12
Annualized Return (%): 15.12
Sharpe Ratio: 4.44
Volatility (%): 30.44
```



Backtest Result:

Prediction based DCA achieved 15.18% return, higher than 14.67% of ADCA, and 15.12 of EDCA, proven that it outer perform than existing strategies.

Conclusion

What i Found:
There are similarity on Trump 1st and 2nd term, including the price trend, pair test of p-values of 0.6 on Trump 1st month in office, 0.4 on Trade war period.

News sentiment and Economic data are similar

What i Found:
Thanks to the accurate prediction on future log return, under the approach of Fine Tuned Bidirectional LSTM (with MSE of 0.05), it successfully beat the result of DCA, EDCA, and ADCA, for 15% of profit.

For future improvement, it is need to test not limited to a monthly bases, but others. Also,a Stop loss is need to be set.

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Thank You